Regulatory Capture via Relative Prices*

Gabriel Cañedo Riedel†
University of California, San Diego

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Abstract

Regulations can increase marginal costs by different proportions across firms. Every price rises, but the least burdened firms raise their prices least. They become relatively cheaper, attract demand, and gain revenue, profit, and market share. I find evidence of this mechanism under the Food Safety Modernization Act (FSMA) using Nielsen scanner data. Large firms supported the act, which disproportionately affected their smaller competitors. They grew in regulated categories but not in similar unaffected placebos. For example, in oil-roasted nuts two firms with a 10:1 revenue ratio before the act ended with a 30:1 ratio afterward, with no effect on the raw-nuts placebo. The general-equilibrium estimates imply a welfare cost of at least $362 million per year, with only the largest sixth to third of firms benefiting. This paper brings Raising Rivals' Costs to differentiated goods and general equilibrium.

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†gcanedoriedel@ucsd.edu. Website: gabriel.cr.

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1 Introduction

Firms seem to lobby for regulations that raise their own costs. The largest food manufacturers supported and lobbied for the Food Safety Modernization Act (FSMA), the most important reform of U.S. food-safety law since 1938. Amazon has publicly campaigned for Congress to double the federal minimum wage. Apple has called for stricter data-privacy regulation, and Meta for content-moderation legislation. Why would firms ask for regulations that make their own operations more costly?

I propose that an increase in marginal costs that differs in proportion across firms is enough for some of them to profit. The mechanism works through *relative prices*. When a regulation raises every firm's cost, every price rises, but the least burdened firms raise their prices least. They become relatively cheaper, attract demand, and gain revenue, profit, and market share while their rivals shrink. I focus on regulations that affect more productive firms proportionally less. I call this *regressive incidence*. Examples include regulations where the most productive firms already operate the required systems, use less of the regulated input, or pay above the proposed minimum wage; in general, regulations the most productive firms are better equipped to meet.

FSMA belongs to this category. Compliance was far cheaper for the largest manufacturers than for their smaller rivals. The act largely codified practices these firms had implemented since the 1990s (food-safety plans that included hazard analysis, environmental monitoring, and hygienic zoning) presumably to meet requirements of national retailers and of EU import rules. For smaller rivals, the same obligations meant new programs and new personnel. Monitoring, testing, sanitizing, and record keeping recur with production, raising marginal costs. Public records indicate that the Grocery Manufacturers Association (GMA), which represents the largest food manufacturers, fought to shape the act so that it was almost costless for them, and opposed exemptions for smaller competitors.[1]

The pattern is not exclusive to the food industry. A higher federal minimum wage would raise Amazon's own costs through suppliers and contractors that provide cleaning and

[1] At the April 24, 2008 House hearing on the first draft of the act, GMA's president urged Congress "to require that every food company has a food safety plan that is based upon food safety risk analysis" (Hearing before the Subcommittee on Health, House Committee on Energy and Commerce, 110th Cong., Serial No. 110-108, statement of Calvin Dooley). In June 2009 its next president testified that "we strongly support proposals to require all food manufacturers to conduct a hazard analysis" (Hearing before the same subcommittee, 111th Cong., Serial No. 111-41, statement of Pamela Bailey). On the implementing rule (78 FR 3646, January 16, 2013), GMA endorsed "regulations on preventive controls that would be essentially cost neutral for food companies that already have advanced food safety systems," objected that the proposal as written "does not accommodate existing systems that are used effectively today," and asked FDA to "keep what works." On the exemption for small firms, it wrote that "food safety practices should be optimized in all food processing operations regardless of size" (GMA, Comments on Docket No. FDA-2011-N-0920, November 22, 2013, pp. 4, 8, 22, and 97).

delivery services at minimum-wage pay. Its exposure would be indirect, however, because its own workers already earn above the proposed minimum wage.[2] Local retailers that pay lower wages would have greater exposure.

A privacy statute would restrict Apple's own data collection and use, but its revenue, unlike that of rivals such as Android,[3] does not rely on the targeted advertising such a statute would restrict.[4] A moderation mandate would subject Meta's own platforms to compliance and reporting obligations. But Meta already operates the industry's largest content-moderation infrastructure,[5] while smaller platforms would have to build and sustain one.[6]

By reallocating sales toward the most productive firms, regressive incidence reshapes the firm-size distribution. The distribution of firm size has grown steadily more unequal for at least a century (Kwon et al., 2024; Covarrubias et al., 2020). The reallocation of sales toward each industry's largest, low-labor-share firms drives labor's declining share (Autor et al., 2020). Market share has shifted toward larger, high-markup firms, driving the rise in the aggregate markup since 1980 (De Loecker et al., 2020). The resulting markup dispersion lowers aggregate productivity and welfare (Edmond et al., 2023). A fatter-tailed size distribution also makes the economy less stable because idiosyncratic shocks to large firms do not average out: shocks to the hundred largest U.S. firms account for one third of output-growth fluctuations (Gabaix, 2011). Firm-size inequality is also wage inequality, as larger firms pay more than small firms (Brown and Medoff, 1989; Bloom et al., 2018). These literatures do not study regulation, but they show why regulations that reshape the firm-size distribution matter.

[2]On October 2, 2018, Amazon raised its own minimum wage to $15 an hour and pledged to lobby Congress to double the federal minimum, then $7.25. For most of Amazon's full-time workers, however, $15 was already the effective wage: the announced raise came with the phase-out of restricted-stock grants and monthly attendance bonuses averaging $1,800–$3,000 a year. Some workers reported that they would earn *less* under the new terms. See "Despite minimum wage increase, some Amazon workers say losing stock options and bonuses means they will make less," *TechCrunch*, October 3, 2018, and "Response to Senator Sanders," Amazon, August 29, 2018.

[3]Android is licensed for free and is funded by targeted advertising that uses the data it collects.

[4]Apple endorsed "comprehensive federal privacy legislation" in Bud Tribble's testimony at the Senate Commerce Committee hearing on Examining Safeguards for Consumer Data Privacy, September 26, 2018. Tim Cook repeated the call to regulators worldwide in his keynote to the International Conference of Data Protection and Privacy Commissioners in Brussels, October 2018, and to Congress in "It's Time for Action on Privacy," *TIME*, January 2019.

[5]The company reported some 30,000 people working on safety and security, roughly half of them reviewing content (Zuckerberg, "A Blueprint for Content Governance and Enforcement," November 2018).

[6]Zuckerberg first called for content rules in "The Internet needs new rules," *Washington Post*, March 30, 2019. He subsequently took the request to Congress twice. At a Senate Commerce Committee hearing devoted to Section 230 on October 28, 2020, he called on Congress to update the law. Before the House Energy and Commerce Committee on March 25, 2021, he proposed conditioning Section 230 liability protection on platforms demonstrating systems to identify and remove unlawful content.

Demand reallocation through relative prices has not been studied in the literature on cost-raising regulation. Regulations that raise marginal costs in different proportions are studied in the homogeneous-good partial-equilibrium framework of Raising Rivals' Costs (Williamson, 1968; Salop and Scheffman, 1983; Fowlie et al., 2016). Homogeneous goods, however, have a single price. In differentiated markets, regulation that affects some firms more than others is studied through fixed costs alone (Rebeyrol, 2023; Macedoni and Weinberger, 2026), but selection through fixed costs leaves relative prices among survivors unchanged.[7] This paper brings Raising Rivals' Costs to differentiated goods and general equilibrium.

I first develop the mechanism in partial equilibrium, for a differentiated-goods industry in monopolistic competition facing consumers with homothetic preferences (Matsuyama and Ushchev, 2017, 2020). The main theorem states that a firm benefits whenever the industry's price aggregator increases proportionally more than its own marginal cost. Under regressive incidence, the most productive firms gain revenue, profit, and market share; the least productive firms shrink; and the size distribution becomes more unequal.

I then specialize to CES preferences to derive the estimating equation that I take to FSMA. Under CES, log revenue is linear in log productivity. I call the slope the *productivity–size gradient*: how steeply firm size increases with productivity. Regressive incidence on marginal costs steepens the gradient, making the industry more unequal; progressive incidence flattens it; proportional incidence and selection through fixed costs leave it unchanged.

I apply the estimating equation to FSMA using Nielsen scanner data from 2011–2019, around the September 2016 compliance date. The analysis focuses on dry, ready-to-eat foods exposed to *Salmonella*. Long considered low-risk because they are dry, these foods came under mandatory preventive controls only after the act. I compare products where FSMA binds with similar exempt products that serve as placebos.

I find evidence of regressive incidence on marginal costs: the gradient steepens where FSMA binds and remains unchanged in the paired placebos. Among nuts, it increases by 0.24 for dry roasters and 0.46 for oil roasters, both of which must control contamination in their own plants. The effect in oil-roasted nuts means that two firms with a 10:1 revenue ratio in 2011 ended up with a ratio of 30:1 after the act. The gradient remains unchanged for handlers of raw nuts, who shift the control to their suppliers.

Among dried fruit, the gradient increases for true dried fruit and remains unchanged for fruit snacks, whose terminal cook removes the need for control. In chocolate, it increases where nut and peanut-butter fillings enter after cocoa sterilization, leaving the firm as the last line of control. It remains unchanged for plain bars, which transfer the control to cocoa

[7] Appendix A organizes this literature.

roasters. I also study categories that lack paired placebos: the gradient increases by 0.10 in peanut butter; by 0.06 in breakfast cereal; and in seasoning and flavored crackers.

I extend the mechanism to general equilibrium in a nested CES economy in which a single industry faces a regulation with regressive incidence. Under the standard assumption that demand across industries is inelastic (Atkeson and Burstein, 2008; Comin et al., 2021; Andreyeva et al., 2010), the mechanism strengthens. This implies that the regulated industry's top firms grow even more than in partial equilibrium.

The estimating equation derived in partial equilibrium holds exactly in the general-equilibrium model. I therefore use its estimates in the general-equilibrium model to quantify the welfare cost of FSMA. With Pareto-distributed marginal costs, the estimated change in the productivity–size gradient corresponds to a shift in the distribution's shape parameter, fattening its tail and concentrating the industry. Across the treated categories, the estimates imply price-index increases of up to ten percent, with only the top sixth to third of firms growing while the rest shrink. The total welfare cost is at least $362 million per year.

Finally, I use the simplest Raising Rivals' Costs model to show how studying FSMA within the prevailing framework can reverse the welfare conclusion. In this model, the leader's market share reflects its productivity advantage over the rest of the industry. The model counts reallocation from less productive firms to the leader as an efficiency gain. Therefore, once the leader's market share exceeds a closed-form threshold, this efficiency gain outweighs the costs of the regulation, and welfare increases. The leader crosses that threshold in four of the products I study.

The paper proceeds as follows. Section 2 develops the mechanism in partial equilibrium and derives the gradient-change estimating equation from a CES specialization. Section 3 presents the FSMA application. Section 4 develops the general-equilibrium extension and the structural quantification. Section 5 examines the conclusions a homogeneous-good model would yield. Section 6 concludes.

2 Regressive Incidence in Partial Equilibrium

I begin in partial equilibrium, holding expenditure and input costs fixed; Section 4 endogenizes them. In a differentiated industry with homothetic single-aggregator demand, a firm's revenue, profit, and market share depend only on its marginal cost relative to the price aggregator. I use this relationship to determine which firms grow under regulation and to compare regressive, proportional, and progressive incidence. A CES specialization then yields an estimating equation that uses changes in the productivity–size gradient to estimate the elasticity of incidence and distinguish the incidence types.

2.1 Environment

Consider a static industry with a unit continuum of differentiated varieties indexed by $\omega \in (0,1)$, each produced by a single firm. Preferences over varieties are homothetic with a single aggregator (HSA), as in Matsuyama and Ushchev (2017, 2020). Demand for variety ω is

$$x_\omega \;=\; \frac{Y}{p_\omega}\, s_\omega\!\left(\frac{p_\omega}{P(\mathbf{p})}\right), \tag{1}$$

where $Y > 0$ is fixed industry expenditure, $s_\omega(\cdot)$ is the share function, and $P(\mathbf{p})$ is the price aggregator.[8] The aggregator is defined implicitly by the adding-up condition

$$\int_0^1 s_\omega\!\left(\frac{p_\omega}{P(\mathbf{p})}\right) d\omega \;=\; 1\,. \tag{2}$$

I assume throughout that the goods are gross substitutes, $s'_\omega(z) < 0$: a firm that raises its relative price loses expenditure share. Under CES, this condition is $\sigma > 1$. The adding-up condition (2) pins down the aggregator and gives it three properties. First, $P(\mathbf{p})$ exists and is unique (Matsuyama and Ushchev, 2017). Second, P is linearly homogeneous: scaling every price and the aggregator by the same factor leaves relative prices and the adding-up integral unchanged, so uniqueness implies $P(\lambda\mathbf{p}) = \lambda\, P(\mathbf{p})$. Third, P is increasing in the price profile.

Firm ω has constant marginal cost $\psi_\omega > 0$ and chooses its price to maximize

$$\pi_\omega \;=\; (p_\omega - \psi_\omega)\, x_\omega \;=\; \left(1 - \frac{\psi_\omega}{p_\omega}\right) Y\, s_\omega\!\left(\frac{p_\omega}{P(\mathbf{p})}\right).$$

[8] I use $P(\mathbf{p})$ for Matsuyama and Ushchev's common price aggregator $A(\mathbf{p})$. It is the single statistic through which rivals' prices enter a firm's demand.

Write $z_\omega \equiv p_\omega / P(\mathbf{p})$ for the relative price. The profit function becomes

$$\pi_\omega \;=\; \left(1 - \frac{\psi_\omega / P(\mathbf{p})}{z_\omega}\right) Y\, s_\omega(z_\omega), \tag{3}$$

so maximizing over price levels is equivalent to maximizing over relative prices. The firm's marginal cost enters the problem only through its *relative marginal cost*, $\psi_\omega / P(\mathbf{p})$. The firm takes the aggregator as given and trades a wider margin against a smaller share. Its first-order condition is

$$\frac{\psi_\omega / P(\mathbf{p})}{z_\omega^2}\, s_\omega(z_\omega) \;+\; \left(1 - \frac{\psi_\omega / P(\mathbf{p})}{z_\omega}\right) s'_\omega(z_\omega) \;=\; 0. \tag{4}$$

The first term captures the gain from a wider margin; the second captures the loss of market share. Write $\zeta_\omega(z) \equiv -\, z\, s'_\omega(z) / s_\omega(z) > 0$ for the elasticity of the share function. Demand (1) is the product of Y/p_ω and the share function, so the demand elasticity perceived by the firm is $1 + \zeta_\omega(z)$, which exceeds one under gross substitutes. Multiplying (4) by $z_\omega / s_\omega(z_\omega)$ and collecting terms gives

$$p^*_\omega \;=\; \mu_\omega\, \psi_\omega, \qquad \mu_\omega \;=\; \frac{1 + \zeta_\omega(z^*_\omega)}{\zeta_\omega(z^*_\omega)} \;>\; 1. \tag{5}$$

The firm therefore charges a finite markup over marginal cost. Under CES, $\zeta = \sigma - 1$ for every variety, so every firm charges $\mu = \sigma/(\sigma - 1)$.

At the equilibrium price profile $\mathbf{p}^* = \boldsymbol{\mu}\boldsymbol{\psi}$, the envelope theorem applied to (3) gives

$$\frac{\partial \pi^*_\omega}{\partial\big(\psi_\omega / P\big)} \;=\; -\,\frac{Y\, s_\omega(z^*_\omega)}{z^*_\omega} \;<\; 0. \tag{6}$$

Equilibrium revenue and market share are also strictly decreasing in ψ_ω / P: the optimal relative price z^*_ω increases with relative marginal cost, while the share function decreases with relative price.[9] Relative marginal cost is therefore the sufficient statistic for a firm's competitive position: it determines its relative price, market share, revenue, and profit.

Without loss of generality, index firms by marginal cost: $\psi_\omega = G^{-1}(\omega)$ for a continuous, strictly increasing cost distribution G on $[\psi_0, \infty)$, with $\psi_0 > 0$. Low ω therefore denotes low

[9] Differentiating equilibrium revenue and market share with respect to relative marginal cost gives

$$\frac{\partial\,[Y\, s_\omega(z^*_\omega)]}{\partial(\psi_\omega / P)} = Y\, s'_\omega(z^*_\omega)\, \frac{\partial z^*_\omega}{\partial(\psi_\omega / P)} < 0,$$
$$\frac{\partial\, s_\omega(z^*_\omega)}{\partial(\psi_\omega / P)} = s'_\omega(z^*_\omega)\, \frac{\partial z^*_\omega}{\partial(\psi_\omega / P)} < 0.$$

cost, and the most productive firms lie near $\omega = 0$.

2.2 Regulations as deformations

Definition 1 (Cost-increasing regulation)**.** *A* regulation *is a change from the marginal-cost profile ψ to ψ^R. It is* cost-increasing *if*

$$\psi^R_\omega \geq \psi_\omega \qquad \text{for every } \omega,$$

with strict inequality on a set of positive measure.

Theorem 1 (Who wins)**.** *Under a deformation $\psi \to \psi^R$, firm ω's equilibrium revenue, profit, and market share strictly increase if and only if its relative marginal cost decreases:*

$$\frac{\psi^R_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi}^R)} < \frac{\psi_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi})}.$$

All three outcomes strictly decrease if the inequality is reversed and remain unchanged if it holds with equality.

Proof in Appendix B.

A firm grows when the regulation raises the price aggregator proportionally more than its own marginal cost. Regressive incidence is sufficient for the most productive firms to satisfy this condition:

Proposition 1 (Regressive incidence)**.** *Consider a cost-increasing regulation with incidence $1 + \varepsilon_\omega \equiv \psi^R_\omega / \psi_\omega$. Suppose incidence is* regressive*:*

$$\varepsilon_\omega \leq \varepsilon_{\omega'} \qquad \text{whenever } \omega < \omega',$$

with strict inequality for at least one pair. Then there exist cutoffs $0 < \bar{\omega} \leq \bar{\omega}' < 1$ such that every firm in $(0, \bar{\omega})$ strictly gains revenue, profit, and market share, while every firm in $(\bar{\omega}', 1)$ strictly loses all three.

Proof in Appendix B.

No firm needs to be fully exempt. It is enough that the most productive firms are the *least* burdened, for example because they already operate the required systems, use less of the regulated input, or pay near the proposed floor.[10]

[10] Appendix C develops two stronger variants of Proposition 1: one in which incidence vanishes at the top of the productivity distribution, and one in which a kink spares the most productive firms.

The argument uses the aggregator's monotonicity and linear homogeneity (Section 2.1). Under regressive incidence, every firm faces at least the incidence of the most productive firms, and some face strictly more. The aggregator therefore increases proportionally more than the most productive firms' marginal costs, lowering their relative costs. Theorem 1 then implies that they grow.

At the other end of the productivity distribution, every firm faces at most the incidence of the least productive firms, and some face strictly less. The aggregator therefore increases proportionally less than the marginal costs of the least productive firms, so their relative costs increase.

Selection through exit further amplifies the gains of the most productive firms.

Corollary 1 (Exit amplifies)**.** *Under the conditions of Proposition 1, suppose firms must pay a fixed operating cost $F > 0$ to remain active and a positive measure of firms exits under the regulation. Then every surviving firm in $(0, \bar{\omega})$ gains strictly more than under the same regulation without exit.*

Proof in Appendix B.

With fewer active varieties, the adding-up condition (2) requires shares to be renormalized over a smaller set. This pushes the aggregator further up, amplifying the relative-cost mechanism. The survivors' gains are net of F, which the regulation leaves unchanged.

2.3 Proportional and progressive incidence

The existing literature on marginal-cost regulation in differentiated markets concludes that these regulations are neutral (Macedoni and Weinberger, 2026; Anouliès, 2017; Rebeyrol, 2023). Proposition 2 shows that these results reflect the particular deformation studied: a proportional increase in a homothetic environment. Neutrality is then a property of that deformation, rather than of marginal-cost regulation itself (Appendix A).

Proposition 2 (Proportional incidence is neutral)**.** *Suppose $\psi_\omega^R = (1+\varepsilon)\psi_\omega$ for every ω, with a common $\varepsilon > 0$. Each firm's relative marginal cost is unchanged:*

$$\frac{\psi_\omega^R}{P(\boldsymbol{\mu}\boldsymbol{\psi}^R)} = \frac{(1+\varepsilon)\psi_\omega}{P((1+\varepsilon)\boldsymbol{\mu}\boldsymbol{\psi})} = \frac{\psi_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi})} \qquad \text{for every } \omega.$$

Every firm's revenue, profit, and market share therefore remain unchanged.

Proof in Appendix B.

An ad valorem tax and a uniform increase in the price of a common input bundle are examples of proportional incidence. Each scales every marginal cost by the same factor. Linear

homogeneity makes the aggregator scale by that factor as well, leaving relative marginal costs unchanged.

Proposition 3 (Progressive incidence)**.** *Consider a cost-increasing regulation with incidence* $1+\varepsilon_\omega \equiv \psi^R_\omega/\psi_\omega$*. Suppose incidence is* progressive*:*

$$\varepsilon_\omega \geq \varepsilon_{\omega'} \qquad \text{whenever } \omega < \omega',$$

with strict inequality for at least one pair. Then there exist cutoffs $0 < \bar{\omega} \leq \bar{\omega}' < 1$ *such that every firm in* $(0, \bar{\omega})$ *strictly loses revenue, profit, and market share, while every firm in* $(\bar{\omega}', 1)$ *strictly gains all three.*

Proof in Appendix B.

Progressive incidence redistributes *against* the most productive firms: they shrink while the least productive firms grow.

Remark 1 (Deregulation). Every statement has a symmetric counterpart for cost-*decreasing* deformations. In particular, removing a regressive incidence regulation strictly harms the most productive firms.[11]

I now specialize to CES to derive the estimating equation used in Section 3.

2.4 A CES specialization

I specialize the demand system in Section 2.1 to CES by setting $s_\omega(z) = z^{1-\sigma}$, with $\sigma > 1$. The adding-up condition (2) becomes $\int_0^1 (p_\omega/P)^{1-\sigma}\, d\omega = 1$, which yields the CES price index

$$P \;=\; \left(\int_0^1 p_\omega^{1-\sigma}\, d\omega\right)^{\frac{1}{1-\sigma}}. \tag{7}$$

As before, I hold expenditure Y and the price of the input bundle fixed; Section 4 endogenizes both. Firm ω produces at constant marginal cost

$$\psi_\omega \;=\; \frac{e}{\varphi_\omega},$$

where e is the price of the input bundle, common to all firms, and φ_ω is the firm's productivity. The share elasticity is constant at $\zeta = \sigma - 1$, so (5) implies that every firm charges the CES

[11] Appendix H examines how far Propositions 1 and 2 extend beyond HSA demand. Maximizing over price levels remains equivalent to maximizing over relative prices in the other homothetic classes of Matsuyama and Ushchev (2017). Proposition 2 holds in all these classes, and Proposition 1 extends under the conditions stated in the appendix. The appendix also considers demand systems outside these classes; without homotheticity, Proposition 2 itself fails.

markup $\mu = \sigma/(\sigma-1)$ over marginal cost. Firm ω's revenue is $r_\omega = Y\, s_\omega(z_\omega) = Y\,(p_\omega/P)^{1-\sigma}$. Taking logs gives

$$\ln r_\omega \;=\; \underbrace{\ln Y + (\sigma-1)\ln\frac{P}{\mu\, e}}_{\text{common to all firms}} \;+\; (\sigma-1)\;\underbrace{\ln\varphi_\omega}_{\text{firm specific}}\quad. \tag{8}$$

Log revenue is linear in log productivity. Expenditure, the price index, the input price, and the markup enter the common intercept; productivity is the only term that varies across firms. I call the slope the *productivity–size gradient*: it measures how steeply firm size increases with productivity. Throughout, I measure this gradient against pre-regulation productivity φ_ω, which I treat as the firm's type. Before regulation, the CES gradient is $\sigma - 1$, the rate at which a productivity difference translates into a size difference.

2.5 What a regulation does to the gradient

Let a regulation multiply firm ω's marginal cost by $1 + \varepsilon_\omega \geq 1$, the firm's incidence. Its marginal cost becomes $(1+\varepsilon_\omega)\, e/\varphi_\omega$, so it produces as if its productivity were $\varphi_\omega^{\text{post}} = \varphi_\omega/(1+\varepsilon_\omega)$. Applying (8) before and after regulation gives

$$\ln r_\omega^{\text{pre}} \;=\; \ln Y + (\sigma-1)\ln\frac{P}{\mu\, e} + (\sigma-1)\ln\varphi_\omega\,, \tag{9}$$

$$\ln r_\omega^{\text{post}} \;=\; \ln Y + (\sigma-1)\ln\frac{P^R}{\mu\, e} + (\sigma-1)\ln\varphi_\omega \;-\; (\sigma-1)\ln(1+\varepsilon_\omega)\,. \tag{10}$$

Define the *elasticity of incidence* with respect to productivity as

$$\tau \;\equiv\; -\frac{d\ln(1+\varepsilon_\omega)}{d\ln\varphi_\omega} \;>\; 0. \tag{11}$$

I assume that τ is constant, so integration gives

$$\ln(1+\varepsilon_\omega) \;=\; c \;-\; \tau\,\ln\varphi_\omega\,, \tag{12}$$

where c is a level common to all firms. Substituting (12) into (10) gives

$$\ln r_\omega^{\text{post}} \;=\; \underbrace{\ln Y + (\sigma-1)\ln\frac{P^R}{\mu\, e} \;-\; (\sigma-1)\,c}_{\text{common to all firms}} \;+\; (\sigma-1)(1+\tau)\;\underbrace{\ln\varphi_\omega}_{\text{firm specific}}\quad. \tag{13}$$

The gradient changes from $\sigma - 1$ before regulation to $(\sigma - 1)(1 + \tau)$ afterward, a change of $(\sigma - 1)\tau$.

Define $\Delta \ln r_\omega = \ln r_\omega^{\text{post}} - \ln r_\omega^{\text{pre}}$. With expenditure and the input price fixed, subtracting (9) from (10) gives

$$\Delta \ln r_\omega \;=\; \underbrace{(\sigma - 1)\big(\Delta \ln P - c\big)}_{\text{common to all firms}} + (\sigma - 1)\,\tau\, \ln \varphi_\omega \,. \tag{14}$$

Under *proportional* incidence, $\tau = 0$ and $\Delta \ln P = c$ (Prop. 2), so $\Delta \ln r_\omega = 0$ and the gradient is unchanged.

Under *selection*, a fixed-cost increase forces the least productive firms to exit. Survivors' marginal costs are unchanged ($c = \tau = 0$), but exit raises the price aggregator, as in Corollary 1. Every survivor therefore gains revenue ($\Delta \ln r_\omega > 0$), while the gradient remains unchanged. Appendix J derives this invariance in two benchmark models: the competitive industry of Hopenhayn (1992) and the selection model of Melitz (2003).

Under *regressive* incidence, $\tau > 0$, and firm ω grows if and only if

$$\tau\, \ln \varphi_\omega \;>\; c - \Delta \ln P \qquad \Longleftrightarrow \qquad \ln(1 + \varepsilon_\omega) \;<\; \Delta \ln P \,. \tag{15}$$

By Proposition 1, this condition holds for the most productive firms, which gain revenue, profit, and market share.[12] Figure 1 illustrates the effect of each regulatory change on the gradient.

2.6 Measuring the change

I do not observe productivity directly, so I use baseline revenue (a firm's size before the estimation window) as a proxy.[13] This model-correct proxy also removes the factor $\sigma - 1$ from

[12] Under CES, the increase in the price aggregator is a share-weighted power mean of the incidences. To first order,

$$\Delta \ln P \simeq \int_0^1 s_\gamma \, \ln(1 + \varepsilon_\gamma)\, d\gamma,$$

where s_γ is the pre-regulation revenue share. A firm grows when its own incidence falls below the industry's share-weighted average:

$$\ln(1 + \varepsilon_\omega) < \int_0^1 s_\gamma \, \ln(1 + \varepsilon_\gamma)\, d\gamma.$$

[13] The positive relationship between firm size and productivity is a robust finding of the empirical productivity literature. Within industries, aggregate productivity exceeds the unweighted firm average precisely because revenue shares and productivity are positively correlated (Olley and Pakes, 1996; Syverson, 2011; Bartelsman et al., 2013). This relationship has been documented in both manufacturing and services across seventeen countries (Berlingieri et al., 2018).

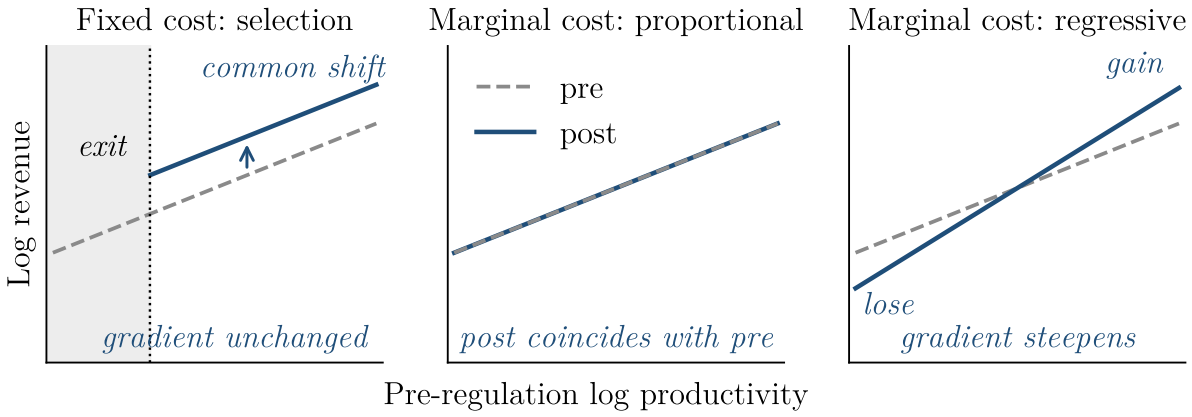


Figure 1: The productivity–size gradient under three regulatory changes. The gradient is the slope of log revenue with respect to pre-regulation log productivity. Dashed grey lines show the relationship before regulation; solid blue lines show it afterward. A fixed-cost increase removes the least productive firms and raises every survivor's revenue by a common factor, leaving the gradient unchanged. Proportional incidence leaves every firm's revenue unchanged. Regressive incidence steepens the gradient.

the estimated change in the gradient, so the coefficient measures the elasticity of incidence directly. By (8), baseline log revenue is an affine transform of log productivity:

$$\ln r_\omega^0 \;=\; A^0 \;+\; (\sigma-1)\ln\varphi_\omega \qquad\Longleftrightarrow\qquad \ln\varphi_\omega \;=\; \frac{\ln r_\omega^0 - A^0}{\sigma-1}, \tag{16}$$

where A^0 is the common baseline intercept. Baseline revenue therefore ranks firms exactly as productivity does, with log differences scaled by $\sigma-1$. Substituting (12) and (16) into (14) gives

$$\Delta\ln r_\omega \;=\; A^\Delta \;+\; \tau\,\ln r_\omega^0, \tag{17}$$

where A^Δ collects A^0, c, and the price aggregator's response.

To obtain the estimating equation, let A^{pre} denote the common terms in (9). Before regulation, substituting (16) into (9) gives

$$\ln r_\omega^{\text{pre}} \;=\; \left(A^{\text{pre}} - A^0\right) \;+\; \ln r_\omega^0 .$$

Pre-period log revenue has unit slope in baseline log revenue; all common terms enter the intercept. After regulation, adding the response in (17) gives

$$\ln r_\omega^{\text{post}} \;=\; \ln r_\omega^{\text{pre}} + \Delta\ln r_\omega \;=\; \left(A^{\text{pre}} - A^0 + A^\Delta\right) \;+\; (1+\tau)\,\ln r_\omega^0 .$$

Let post_t indicate the post-regulation period. Combining the pre- and post-regulation expressions gives

$$\begin{aligned}\ln r_{\omega t} &= (1-\text{post}_t)\ln r_\omega^{\text{pre}} + \text{post}_t \ln r_\omega^{\text{post}} \\ &= \left(A^{\text{pre}} - A^0\right) + A^\Delta \cdot \text{post}_t + \ln r_\omega^0 + \tau\left(\ln r_\omega^0 \times \text{post}_t\right).\end{aligned}$$

The terms that vary only by period form the time effects, $\delta_t = \left(A^{\text{pre}} - A^0\right) + A^\Delta \cdot \text{post}_t$. The baseline term varies only by firm and forms the firm effect, $\delta_\omega = \ln r_\omega^0$.[14] Writing β for the coefficient on the interaction gives

$$\ln r_{\omega t} = \beta\left(\ln r_\omega^0 \times \text{post}_t\right) + \delta_\omega + \delta_t, \qquad \beta = \tau.$$

Exponentiating yields the estimating equation:

$$r_{\omega t} = \exp\Big\{\beta\left(\ln r_\omega^0 \times \text{post}_t\right) + \delta_\omega + \delta_t\Big\}. \tag{18}$$

The time effects absorb any aggregate changes in expenditure Y or the input price e. I hold both fixed here for exposition; Appendix I allows them to vary and derives the same estimating equation.

The same coefficient follows directly from the cross-firm slope in (17):

$$\beta = \frac{\text{Cov}\left(\Delta \ln r_\omega,\, \ln r_\omega^0\right)}{\text{Var}\left(\ln r_\omega^0\right)} = \frac{(\sigma-1)\,\tau \cdot (\sigma-1)\,\text{Var}(\ln \varphi_\omega)}{(\sigma-1)^2\,\text{Var}(\ln \varphi_\omega)} = \tau\,.$$

The sign of β provides a test of the type of incidence:

$$\begin{gathered}\text{regressive incidence} \Rightarrow \beta > 0 \qquad \text{progressive incidence} \Rightarrow \beta < 0 \\ \text{proportional incidence or selection through fixed costs} \Rightarrow \beta = 0.\end{gathered}$$

The incidence test extends beyond CES. CES buys the linearity that makes β equal to the elasticity of incidence τ.

[14] This decomposition is unique up to an additive constant.

3 The Application: FSMA

I estimate the elasticity of incidence for the Food Safety Modernization Act.

3.1 The regulation

The Food Safety Modernization Act, signed in January 2011 and finalized in September 2015, is the largest reform of U.S. food-safety law since 1938. Its justification came from the 2008–09 outbreak at the Peanut Corporation of America (PCA), whose *Salmonella*-contaminated peanut paste reached hundreds of products downstream. The outbreak sickened more than seven hundred people, killed nine, and led to criminal convictions.

Yet the resulting regulation would not have prevented the outbreak. PCA already carried out the monitoring that the act later made mandatory. The outbreak originated at its plant in Blakely, Georgia, which held a "superior" rating from its private auditor the year before. Trial records showed that, on at least a dozen occasions in 2007–08, PCA shipped lots under falsified certificates of analysis after positive *Salmonella* tests. The convictions came under statutes that predated FSMA.

The act's centerpiece was the Preventive Controls for Human Food rule (21 CFR Part 117). The rule almost entirely codified practices already used by the largest manufacturers: a written hazard analysis, monitoring of the production environment (§117.165), hygienic zoning, and supplier verification. Compliance began in September 2016, the event studied here.

I focus on dry, low-water-activity, ready-to-eat foods exposed to *Salmonella*, the pathogen behind the outbreak that drove the reform. These foods had long been treated as low-risk because they are dry and came under mandatory preventive controls only after the act. My design compares products within a category: they share the shelf, and face similar supply and demand shocks, but bear different regulatory burdens under FSMA.

Within a category such as peanut butter, nuts, or chocolate bars, FSMA binds when the plant has no *Salmonella* "kill step" (such as frying). It also binds when a kill step is followed by exposed handling (for example, cooling and seasoning before packing) that carries a risk of recontamination. FSMA does not bind when a *Salmonella* kill step occurs at the end of the production line, so no monitoring is needed. It also does not bind when responsibility for contamination monitoring can be shifted upstream, for example to the farmer supplying the processing plant.

The act's obligations recur with every production run. Environmental monitoring (§117.165) requires swabbing and testing the processing environment wherever exposed product travels. A positive test triggers corrective action, re-sanitation, and retesting (§117.150). Hygienic

zoning must be maintained on every shift as people and materials move between the raw and exposed sides of the plant. Each duty generates records that must be kept and reviewed (§117.190). The resulting work hours, tests, and paperwork increase with output, giving the regulation a marginal-cost incidence.

Regressive incidence was not an accident. The large branded manufacturers that lobbied for and supported the act had operated HACCP-style food-safety systems (hazard analysis, monitoring, and documentation) since the 1990s, as these systems were a precondition for exporting to the European Union and Japan. For these firms, the Preventive Controls rule largely codified existing practice: a hazard analysis they had already written and monitoring they already ran, at near-zero incremental cost. For facilities without prior programs, the same text meant new programs, new personnel, and a higher cost per unit of output. Compliance was staggered by size, beginning in September 2016 for the largest firms and in 2017 and 2018 for smaller tiers. Very small businesses, most of them organic producers, were largely exempt.

3.2 Data

I use the 2011–2019 Nielsen Retail Scanner files from the Kilts Center at the University of Chicago. Participating stores report weekly revenue and units for every product sold, identified by its Universal Product Code (UPC), a twelve-digit barcode. The stores span grocery, drug, mass-merchandise, and convenience channels, which I pool. The panel covers more than half of sales in U.S. grocery and drug stores and about a third of sales at mass merchandisers. Walmart and Costco do not participate. Supermarkets account for roughly 90% of revenue in the categories studied here.

Nielsen assigns each UPC to a product module, its most detailed product category, such as peanut butter, cold cereal, or seasoning. The categories in this paper are modules or, when a module combines distinct markets, partitions of a module (Section 3.3). Each UPC carries a brand, which I link to its parent company through an ownership crosswalk constructed from company ownership records as of 2016.[15] Fixing this mapping in 2016 prevents later acquisitions from reassigning revenue across firms. I exclude store brand UPCs because they do not have an identifiable or unique producer.

I retain stores present in every year of the panel: 27,215 of 55,966. This keeps the set of stores constant before and after the event. I then retain only retailers that report without interruption: 67 of 88, accounting for 99% of revenue. Table 1 traces these restrictions.

The unit of observation is the firm–county–quarter. I map weekly revenue by UPC and

[15]The crosswalk links about 2,700 brands to roughly 500 parent companies across the categories studied.

	Stores	Retailers
Stores active in any year, 2012–2019	55,966	199
Stores active in every year	27,215	88
Retailers reporting in all 32 quarters (99% of revenue)	26,966	67

Table 1: Store sample construction. A store is active in a year if it records sales that year. Retailer counts give the number of retailers operating the stores in each row.

store to parent firms through the brand identifiers and the 2016 ownership crosswalk. I then sum revenue across each firm's UPCs, stores within each county, and weeks within each quarter.

I retain a firm–county–quarter cell with zero revenue if the firm has previously sold in that county and still sells elsewhere nationally. I drop quarters in which the firm sells nowhere nationally. The share of zero-revenue cells ranges from 5% to 33% across categories; the Poisson estimator in Section 3.4 accommodates these zeros. Baseline size is the firm's 2011 revenue in the category (Section 3.4). Table 2 summarizes each category: the number of firms and observations, the share of zero-revenue cells, the leading firm's revenue share, and category revenue.

Product	Firms	Obs.	Zero share	Leader share	Revenue 2016 ($M)
Nuts, oil-roasted	24	518,392	0.14	0.45	142
Nuts, dry-roasted	34	604,924	0.21	0.64	273
Nuts, raw	36	380,619	0.24	0.39	136
Dried fruit, true-dried	20	424,678	0.29	0.42	96
Dried fruit, fruit snacks	18	516,413	0.20	0.43	225
Peanut butter, creamy	14	421,110	0.19	0.57	339
Peanut butter, chunky	12	284,799	0.21	0.47	77
Chocolate, nut/PB	35	1,196,003	0.23	0.38	868
Chocolate, plain	27	783,313	0.33	0.46	536
Granola bars, w/ inclusions	15	441,082	0.13	0.40	240
Granola bars, plain	10	339,588	0.18	0.70	120
Granola cereal	12	366,194	0.17	0.31	77
Cereal	25	615,285	0.05	0.37	2,042
Seasoning	35	967,067	0.15	0.56	385
Crackers, flavored	23	719,094	0.15	0.63	491

Table 2: Summary statistics by product. Firm and observation counts refer to the estimation sample: firms above the size floor and firm–county–quarter cells excluding 2016. The zero share is the fraction of cells with zero revenue. The leader share is the largest firm's share of 2016 revenue among sampled firms. Revenue is the sampled firms' 2016 revenue in the balanced-store panel.

3.3 Sample selection

I select products using three criteria:

1. *Product characteristics.* The product must be dry, low-water-activity, ready-to-eat, and in the *Salmonella* low-moisture hazard class, with no terminal kill step. Frying, popping, and cooking sterilize the product at the end of the production line; such categories are excluded or serve as placebos when they can be paired with a treated category.

2. *Market coherence.* The category must form a single competitive market. Several Nielsen modules pool non-substitutes into administrative assortments. Some have no clear partition into markets of substitutes. In other cases, a coherent partition is possible, but the resulting markets contain too few firms to satisfy the firm-count requirement below.

3. *Gradient estimability.* The category must have sufficient firms and size dispersion among them, given the revenue floor of \$1M (see Section 3.4). For example, I exclude categories with nine or fewer firms or with one brand accounting for 80% or more of revenue.

These criteria exclude toaster pastries and most cracker modules, which have fewer than six firms, and sandwich cookies, where a single firm accounts for roughly 83% of revenue. They also exclude administrative residuals that pool non-substitutes, such as non-chocolate candy (from hard candy to gummy bears) and unclassified cookies (from chocolate chip to shortbread). Fifteen products meet the criteria; Table 2 lists them.

3.4 Estimation

Specification. I estimate (18) by Poisson pseudo-maximum-likelihood using firm × county × time revenue (Santos Silva and Tenreyro, 2006):

$$r_{ict} \;=\; \exp\Big\{\, \beta \left(\ln \text{size}_i^{2011} \times \text{post}_t\right) \;+\; \delta_{ic} \;+\; \delta_{st} \,\Big\} \cdot \eta_{ict}, \qquad \mathbb{E}\big[\eta_{ict} \,|\, \cdot\,\big] = 1.$$

Here, i indexes firms, c counties, and t quarters; s denotes the state containing county c. The specification includes firm-by-county fixed effects δ_{ic} and state-by-quarter fixed effects δ_{st}. The coefficient β is the change in the productivity–size gradient at the event (Section 2.6). The estimator is consistent under the displayed moment condition, with no other assumption on the distribution of revenue. The condition permits heteroskedastic or serially correlated deviations. I conduct inference using a wild cluster bootstrap with clustering by firm.

Baseline size. Baseline size is the firm's 2011 revenue within category, measured in the calendar year before the estimation panel. This timing prevents a mechanical link between the treatment variable and outcomes. A \$1M floor on 2011 size guarantees at least two years of compliance: every sampled firm faced compliance by September 2017 at the latest. Compliance began in September 2018 for firms between \$500k and \$1M, while firms below \$500k fall under the "very small businesses" exemption. In categories with very few firms, I lower the floor to \$500k if doing so raises the firm count above nine.

Timing. I use calendar-year bins, with 2015 as the reference year. Compliance began in September 2016, so 2016 contains eight months before compliance and four months after. All difference-in-differences and trend-break estimation therefore uses a donut design that *drops 2016*, with 2012–2015 as the pre period and 2017–2019 as the post period. The event studies still display the 2016 coefficient.

Anticipation. The final rule was published in September 2015 and received wide coverage in the trade press.[16] Its controls take time to implement, including supply-chain changes to accommodate monitoring and hygienic re-zoning of existing production lines. Firms that began preparing to comply when the rule was published would show responses before the September 2016 compliance date. This might explain why effects begin to emerge in 2015 for most products.

Pre-trends. For products with flat pre-trends, I report the unadjusted event study and summarize it with the donut level difference-in-differences. For products with linear pre-trends, I apply the linear trend extrapolation of Dobkin et al. (2018) and Freyaldenhoven et al. (2026). I fit a linear trend to the pre-period coefficients, extrapolate it over the full window, and subtract it from every coefficient, before and after the event. When anticipation makes the 2015 coefficient depart from the 2012–2014 trend, I remove 2015 from the trend fit. This applies to seasoning and granola bars. For cereal, the 2012 coefficient departs from the 2013–2015 trend, so I remove it from the trend fit. The adjusted path is therefore net of the pre-trend's counterfactual continuation. The adjustment also affects the standard errors mechanically. They are small in the pre period, where the trend is fitted, and grow with event time in the post period, where it is extrapolated.

Two estimands summarize these products. The donut *trend break* measures the change in the gradient per year at the event. The trend-adjusted *post average* measures the post-period level net of the extrapolated drift and is the analogue of the donut difference-in-differences for these products. Its standard error comes from the delta method.

[16] E.g., Lydia Zuraw, "FDA Finalizes FSMA Preventive Controls Rules For Human, Animal Foods," *Food Safety News*, September 10, 2015; "F.D.A. finalizes F.S.M.A. preventive controls rules," *Food Business News*, September 11, 2015.

3.5 Results

I organize the results around comparisons within categories: similar products that share shelf space, and face similar supply and demand shocks, but bear different compliance burdens.

Product	Role	$\hat{\beta}$ (level DiD)	s.e.	Obs.	Firms
Nuts, oil-roasted	treated	$+0.456^{***}$	(0.092)	518,392	24
Nuts, dry-roasted	treated	$+0.238^{***}$	(0.077)	604,924	34
Nuts, raw	placebo	-0.075	(0.106)	380,619	36
Dried fruit, true-dried	treated	$+0.292^{***}$	(0.096)	424,678	20
Dried fruit, fruit snacks	placebo	-0.059	(0.065)	516,413	18
Peanut butter, creamy	treated	$+0.103^{***}$	(0.039)	421,110	14
Peanut butter, chunky	treated	$+0.097^{**}$	(0.050)	284,799	12

Table 3: FSMA and the productivity–size gradient: level estimates. The table reports donut level difference-in-differences estimates for products with flat pre-trends. $\hat{\beta}$ measures the post-2016 change in the gradient per log point of 2011 firm revenue within category, the productivity proxy. The pre period is 2012–2015 and the post period is 2017–2019; 2016 is excluded. Estimation uses Poisson PML with firm×county and state×quarter fixed effects. Inference uses a wild cluster bootstrap by firm. Standard errors are in parentheses. The size floor is \$1M. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Nuts. Incidence depends on whether *Salmonella* control is the facility's own duty under §117.165 and cannot be shifted upstream. A roaster has a *Salmonella* kill step but handles the exposed product *afterward*: the nuts must cool before salting or seasoning, and packing. The roaster must therefore control recontamination in its own plant. A handler of *raw* nuts shifts control to its supplier instead. This predicts a change in the gradient for roasted nuts and none for raw nuts. The estimates in Table 3 are +0.238 for dry-roasted nuts and +0.456 for oil-roasted nuts per log point of baseline size. The raw-nut estimate is −0.075, indistinguishable from zero. For oil-roasted nuts, the estimated effect implies that two firms with a ten-to-one sales ratio in 2011 ended the post period with a ratio of nearly thirty to one.

Dried fruit. True dried fruit requires monitoring throughout production. Fruit snacks have no monitoring obligation: they are cooked confections with a terminal cooking step followed immediately by packaging. The estimates are +0.292 for true dried fruit and −0.059 for fruit snacks, the latter statistically indistinguishable from zero.

Peanut butter. Peanut butter is the category that prompted the statute. The estimates are +0.103 for creamy peanut butter and +0.097 for chunky peanut butter, both statistically significant. No paired placebo is available.

Chocolate. Chocolate is the first paired category for which I remove a linear pre-trend.

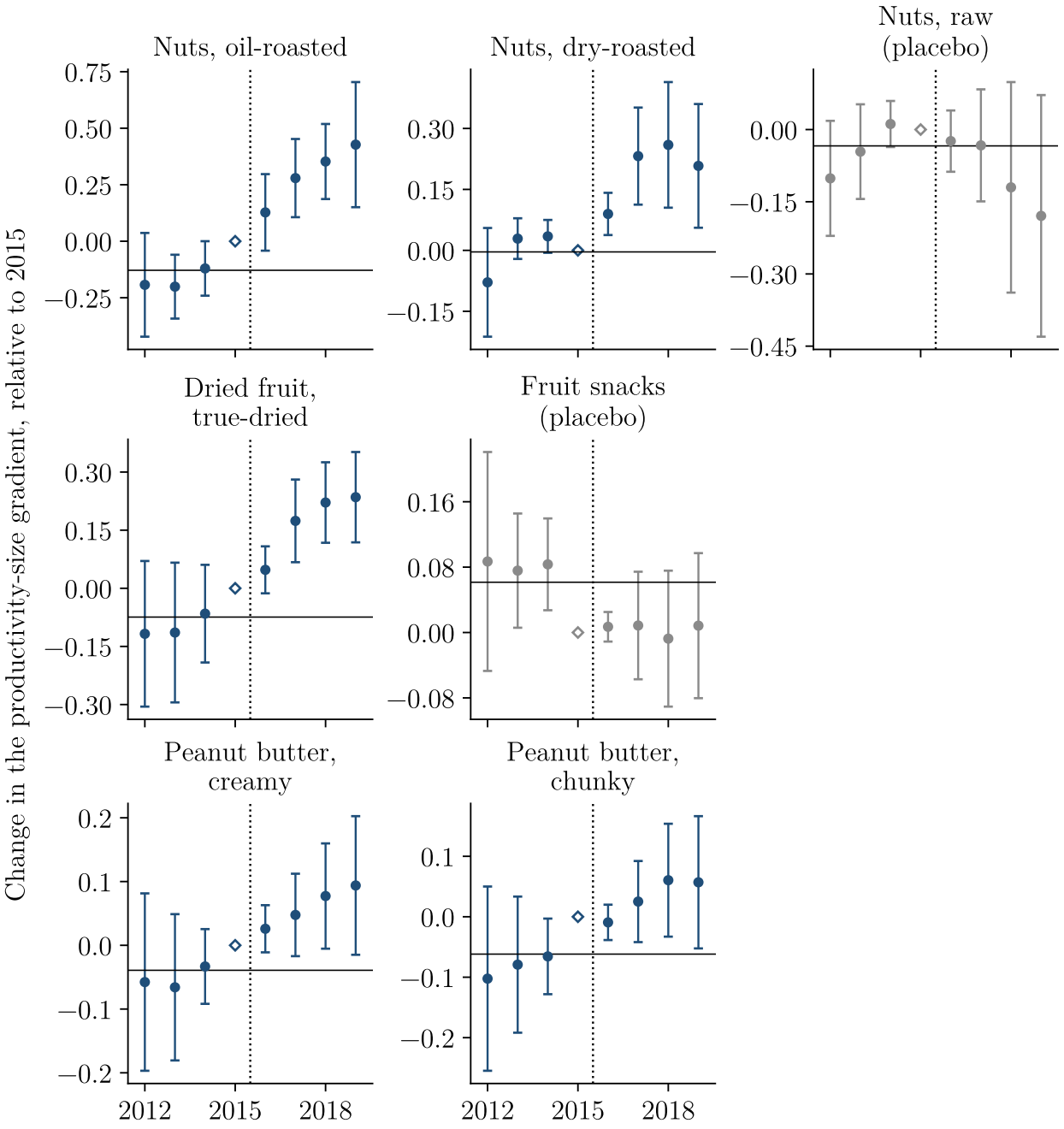


Figure 2: FSMA and the productivity–size gradient: event studies for products with flat pre-trends. The figure reports calendar-year changes in the gradient for the products in Table 3, with 95% confidence intervals. The reference year is 2015, marked by an open diamond. The solid line shows the pre-period average, 2012–2015; the dotted line marks the end of the pre period. Compliance began in September 2016. The 2016 coefficient is plotted; 2016 is excluded from the corresponding donut estimation.

I compare varieties with inclusions (treated) with plain chocolate (placebo). An inclusion is an exposed, ready-to-eat ingredient added *after* the chocolate's own kill step. The man-

Product	Role	Trend break (/yr)	s.e.	Post avg.	Obs.	Firms
Chocolate, nut/PB	treated	$+0.014^{*}$	(0.008)	+0.039	1,041,346	35
Chocolate, plain	placebo	−0.011	(0.026)	−0.025	681,387	27
Granola bars, w/ inclusions†	treated null	+0.035	(0.041)	$+0.348^{*}$	382,778	15
Granola bars, plain†	placebo	−0.046	(0.041)	−0.073	296,845	10
Granola cereal	treated	$+0.112^{***}$	(0.032)	$+0.256^{**}$	317,246	12
Cereal	treated	$+0.036^{***}$	(0.008)	$+0.064^{***}$	549,911	25
Seasoning	treated	$+0.015^{***}$	(0.006)	$+0.049^{***}$	840,400	35
Crackers, flavored	treated	$+0.024^{**}$	(0.012)	+0.027	621,482	23

Table 4: FSMA and the productivity–size gradient: trend-adjusted estimates. The table reports donut trend breaks and trend-adjusted post averages for products with linear pre-trends, after the pre-trend removal of Freyaldenhoven et al. (2026). The trend break measures the change in the gradient per year at the event. The post average measures the post-period level net of the extrapolated pre-trend. Baseline firm size is 2011 revenue within category. The pre period is 2012–2015 and the post period is 2017–2019; 2016 is excluded. Estimation uses Poisson PML with firm×county and state×quarter fixed effects. Inference uses a wild cluster bootstrap by firm. Standard errors in parentheses accompany the trend-break estimates. The size floor is \$1M, except for products marked †, which use a \$500k floor to retain more than nine firms. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

ufacturer is therefore responsible for handling the exposed inclusion in its own plant. Plain chocolate's kill step occurs upstream at cocoa roasting, after which processing is enclosed, with little exposed handling. Chocolate with inclusions shows a marginally significant post-2016 steepening of the gradient (+0.014 per year, $p = 0.08$). Plain chocolate shows no significant change (−0.011, $p = 0.68$; Table 4).

Granola bars: a treated null. I compare granola bars with nut, peanut-butter, fruit, or chocolate inclusions with a plain oats-and-honey placebo. The treated bars carry the same inherited obligation as chocolate with inclusions. The treated event study increases after the event, in the direction predicted by the mechanism. The trend-break estimate is statistically indistinguishable from zero (+0.035, $p = 0.39$), while the trend-adjusted post average is significant (+0.348, $p = 0.06$).

Other products with linear pre-trends. The gradient increases year after year throughout the post period. The estimated increases per log point of baseline size per year are +0.036 for cereal, +0.112 for granola cereal, +0.015 for seasoning, and +0.024 for flavored crackers. All four estimates are statistically significant (Table 4).

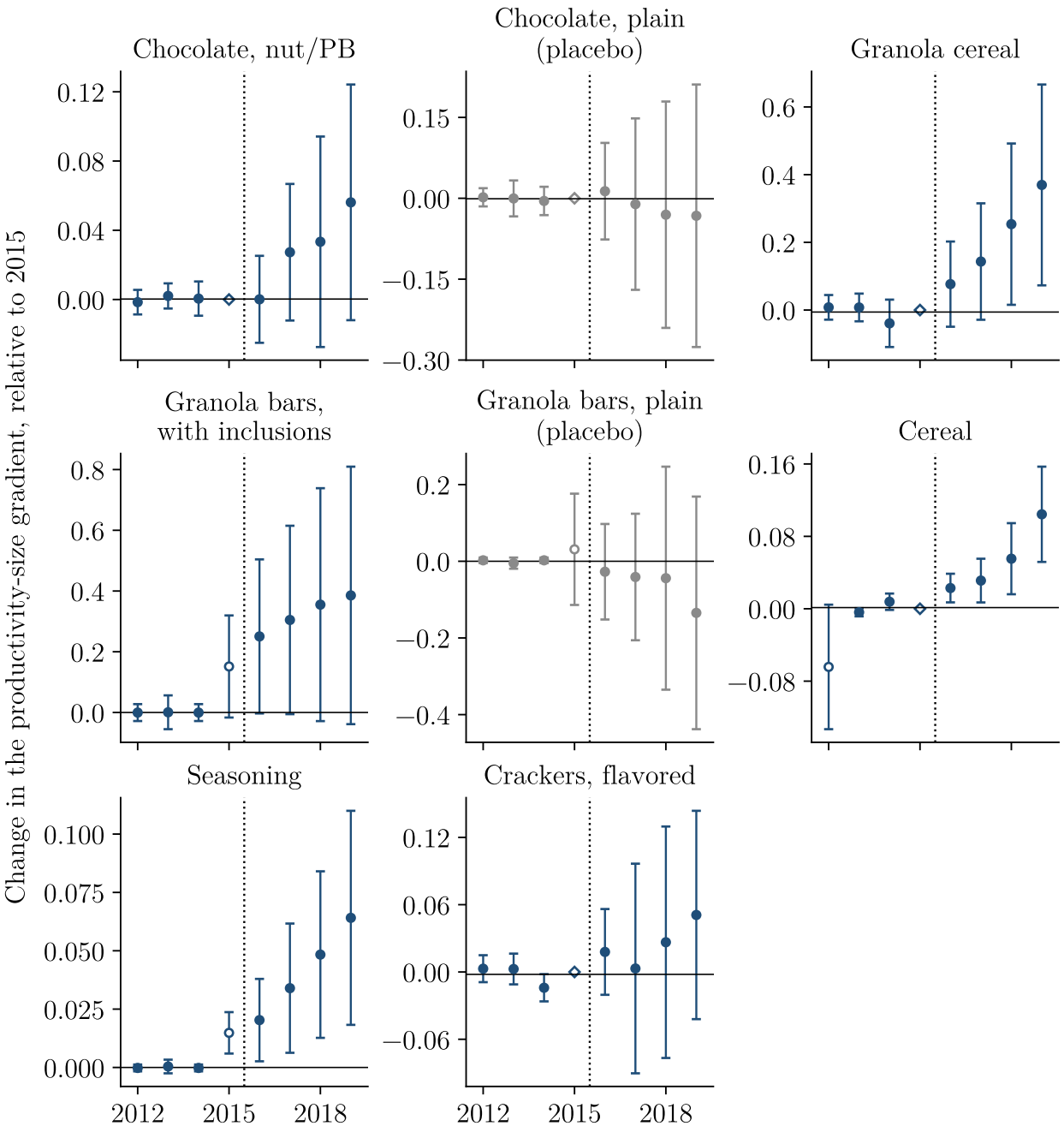


Figure 3: FSMA and the productivity–size gradient: event studies after linear pre-trend removal. The figure reports calendar-year estimates for the products in Table 4, with 95% confidence intervals. The trend fits exclude 2012 for cereal and 2015 for seasoning and granola bars; these years remain plotted with open markers. The reference year is 2015, marked by an open diamond. The solid line shows the pre-period average, 2012–2015; the dotted line marks the end of the pre period. Compliance began in September 2016. The 2016 coefficient is plotted; 2016 is excluded from the corresponding donut estimation.

3.6 The politics

The Grocery Manufacturers Association (GMA) supported the bill throughout its passage and filed forty lobbying disclosures. Its members included General Mills and Kellogg (cereal); Mars, Hershey, and Nestlé (chocolate); J.M. Smucker (peanut butter); McCormick (seasoning); and Mondelēz and Kraft (crackers). At the signing, the GMA celebrated FSMA as "the most comprehensive reform of our nation's food safety laws in more than 70 years" and said it would "help restore the public's faith in the safety and security of the food supply."[17] The GMA's board chairman said the same day: "I am proud of the food industry for its support of landmark food safety legislation."

Opposition came from the other end of the size distribution. Very small producers argued that the burden would be insurmountable. By their own calculations, compliance would consume 4–6% of a small operation's gross sales, more than half its profits at the sector's typical margins.[18] The Organic Consumers Association's grassroots campaign delayed the Senate vote. The Tester amendment resolved the dispute by exempting "qualified facilities." These are facilities with average annual sales below $500,000 that make the majority of their sales directly to consumers or to restaurants and retailers in the same state or within 275 miles (21 U.S.C. §350g(l)).

Lobbying disclosures on FSMA (LobbyView; Huneeus and Kim, 2024) also document activity by individual large firms and industry associations. *Blue Diamond Growers*, the dominant almond processor, filed fifty disclosures in the category where I estimate the roasted-nut gradient. The *American Peanut Shellers Association* filed forty-one in peanut butter, the act's origin category. In chocolate, *Mars*, *Nestlé*, and *Hershey* filed forty-four, thirty-seven, and twenty-eight disclosures, respectively. In cereal, where the gradient also increases, *General Mills*, *Kellogg*, and *Post* also lobbied on the bill. The *American Bakers Association* filed fifty-six disclosures across the baked categories.

The regulation that passed would not have prevented the *Salmonella* outbreak used to justify it. It codified the practices of the industry's largest firms and was backed by their association, the GMA. The estimated changes in the gradient show that its incidence was regressive. Under the mechanism developed here, regressive incidence is sufficient for the most productive firms to gain revenue, profit, and market share. Section 4 shows that this result extends to general equilibrium.

[17] Pamela Bailey, GMA president and CEO, made these remarks on the signing of FSMA, as quoted in Helena Bottemiller, "Historic Food Safety Bill Signed into Law," *Food Safety News*, January 4, 2011.

[18] National Sustainable Agriculture Coalition, "Smaller Farms Likely to Face Higher Food Safety Compliance Costs," and companion FSMA cost analyses. These analyses put compliance costs at 4–6% of gross sales for operations with sales below $500,000, against national average net farm income near 10% of sales—"more than half" of small-operation profits. The coalition also filed thirty-five FSMA lobbying disclosures.

4 The General Equilibrium Closure

Section 2 developed the mechanism in partial equilibrium under homothetic single-aggregator preferences. I now embed it in a nested CES economy. With an outer-nest elasticity $\sigma_0 \leq 1$, a regulation with regressive incidence in a single industry makes its most productive firms gain even more revenue and profit than in partial equilibrium. This elasticity restriction is standard in quantitative models and supported by empirical estimates (Atkeson and Burstein, 2008; Comin et al., 2021; Andreyeva et al., 2010). I use this setting and the estimated elasticity of incidence from the FSMA application to measure the regulation's welfare cost. When demand across industries is elastic, the mechanism weakens, but the most productive firms still grow if varieties substitute more easily within the industry than across industries.

4.1 Nested CES

A representative household supplies L units of labor inelastically and owns all firms. Its preferences over industries $h = 1, \ldots, H$ have an outer CES nest with elasticity σ_0 and positive weights α_h. Within each industry h, preferences over varieties are CES with elasticity $\sigma_h > 1$. Labor is the only factor, and the wage $w = 1$ is the numeraire. The household spends its income Y, which consists of wages and profits. The outer nest determines industry expenditure shares and the aggregate price index:

$$S_h \;=\; \frac{\alpha_h P_h^{1-\sigma_0}}{\sum_k \alpha_k P_k^{1-\sigma_0}}, \qquad P_0 \;=\; \left(\sum\nolimits_h \alpha_h P_h^{1-\sigma_0}\right)^{\frac{1}{1-\sigma_0}}, \tag{19}$$

where S_h is industry h's expenditure share, P_h its CES price index, and P_0 the economy's price index. A firm in industry h charging price p earns revenue $r = (p/P_h)^{1-\sigma_h} S_h Y$.

The regulated industry g is the CES industry of Section 2.4, with elasticity σ_g. Index its firms by pre-regulation productivity φ. A firm with productivity φ uses $1/\varphi$ units of labor per unit of output, so its marginal cost is $\psi = w/\varphi \in [\psi_{\min}, \infty)$. The mass of firms is fixed, and every firm charges the markup $\mu_g = \sigma_g/(\sigma_g - 1)$.

The regulation requires compliance labor. A firm producing q hires production labor ℓ_p and compliance labor ℓ_c in the same labor market at wage w. Production satisfies $q = \varphi\,\ell_p$, and compliance labor is proportional to production labor: $\ell_c = \varepsilon(\psi)\,\ell_p$. Examples include monitoring, sanitizing, and record keeping, as in FSMA. The wage bill is therefore $w(\ell_p + \ell_c) = (1 + \varepsilon(\psi))\, w\, q/\varphi$. The labor requirement increases to $(1 + \varepsilon(\psi))/\varphi$ per unit of output, and marginal cost becomes

$$\psi^R \;=\; (1 + \varepsilon(\psi))\,\frac{w}{\varphi} \;=\; (1 + \varepsilon(\psi))\,\psi.$$

Incidence is regressive: ε is continuous and strictly increasing in ψ, with $\varepsilon(\psi_{\min}) = \varepsilon_0 \geq 0$.

Assumption 1 (Inelastic outer nest)**.** *The elasticity of substitution across industries satisfies* $\sigma_0 \leq 1$.

With $\sigma_0 < 1$, households treat industries as complements. The Cobb–Douglas boundary, $\sigma_0 = 1$, is the workhorse assumption of the quantitative literature.[19]

4.2 The industry's expenditure grows

The regulation changes only industry g's price index. Before regulation,

$$P_g \;=\; \Big(\int (\mu_g\,\psi)^{1-\sigma_g}\,dG(\psi)\Big)^{1/(1-\sigma_g)}.$$

Multiplying each firm's marginal cost by its incidence gives

$$\begin{aligned}\Big(\frac{P_g^R}{P_g}\Big)^{1-\sigma_g} &= \frac{\int_{\psi_{\min}}^{\infty} \psi^{1-\sigma_g}\,\big(1+\varepsilon(\psi)\big)^{1-\sigma_g}\,dG(\psi)}{\int_{\psi_{\min}}^{\infty} \psi'^{\,1-\sigma_g}\,dG(\psi')} \\ &= \int_{\psi_{\min}}^{\infty} s(\psi)\,\big(1+\varepsilon(\psi)\big)^{1-\sigma_g}\,dG(\psi),\end{aligned} \tag{20}$$

where $s(\psi) = \psi^{1-\sigma_g}\big/\int \psi'^{\,1-\sigma_g}\,dG(\psi')$ is a firm's pre-regulation revenue relative to the industry average. The weights $s(\psi)\,dG(\psi)$ therefore integrate to one. By Jensen's inequality, $\Delta \ln P_g > \ln(1+\varepsilon_0)$: the price index increases by more than the incidence of the most productive firms.

Define $\nu \equiv (P_g^R/P_g)^{1-\sigma_0} \geq 1$ as the factor by which industry g's allocation weight, $\alpha_g P_g^{1-\sigma_0}$, changes.[20] Only this term changes in the sum defining $P_0^{1-\sigma_0}$. Dividing the post-regulation sum by the pre-regulation sum expresses each unaffected industry's term as its

[19] Across broad sectors, Comin et al. (2021) estimate the price elasticity of their demand system at 0.26 using U.S. household data and 0.57 using a panel of 39 countries. Herrendorf et al. (2013) use a benchmark elasticity of 0.9. Common quantitative specifications combine a Cobb–Douglas outer nest across industries with CES preferences within industries (Atkeson and Burstein, 2008; Hsieh and Klenow, 2009; Arkolakis et al., 2012). Within food, a review of 160 U.S. studies reports mean across-category price elasticities of 0.27–0.81 (Andreyeva et al., 2010).

[20] The allocation weight of industry g is $\alpha_g P_g^{1-\sigma_0}$, so

$$\frac{\alpha_g (P_g^R)^{1-\sigma_0}}{\alpha_g P_g^{1-\sigma_0}} = \Big(\frac{P_g^R}{P_g}\Big)^{1-\sigma_0} = \nu.$$

pre-regulation expenditure share S_h, by (19):

$$D \;\equiv\; \Big(\frac{P_0^R}{P_0}\Big)^{1-\sigma_0} \;=\; \sum_{h\neq g} S_h + S_g\,\nu \;=\; 1 - S_g + S_g\,\nu \;\geq\; 1. \tag{21}$$

Thus, D is the factor by which the sum of allocation weights, $P_0^{1-\sigma_0}$, changes. It strictly exceeds one when $\sigma_0 < 1$. Substituting P_g^R into (19) and using $P_h^R = P_h$ for every $h \neq g$ gives

$$\begin{aligned} S_g^R &= \frac{\alpha_g\,(P_g^R)^{1-\sigma_0}}{\sum_k \alpha_k\,(P_k^R)^{1-\sigma_0}} = \frac{\nu\,\alpha_g\,P_g^{1-\sigma_0}}{D\,P_0^{1-\sigma_0}} = \frac{S_g\,\nu}{D}, \\ \Delta \ln S_g &= \ln(\nu/D) \geq 0. \end{aligned} \tag{22}$$

The regulated industry's expenditure share increases because $\nu - D = \nu - 1 - S_g(\nu - 1) = (1 - S_g)(\nu - 1) \geq 0$.

Household income is wages plus profits. Under CES, industry h's profit margin (the share of revenue paid out as profit) is $1/\sigma_h$. Write $m_g \equiv 1/\sigma_g$ for the regulated industry's margin and define the expenditure-weighted average margin outside g as

$$m_{-g} \;\equiv\; \sum_{h\neq g} \frac{S_h}{1 - S_g}\,\frac{1}{\sigma_h}.$$

The regulation leaves m_{-g} unchanged: the expenditure shares of all industries $h \neq g$ scale by the same factor, so the weights $S_h/(1 - S_g)$ remain constant. Income therefore satisfies

$$\begin{aligned} Y &= \frac{wL}{1-\kappa}, \\ \kappa &\equiv S_g\,m_g + (1 - S_g)\,m_{-g}, \\ \kappa^R - \kappa &= \big(S_g^R - S_g\big)\big(m_g - m_{-g}\big). \end{aligned} \tag{23}$$

Here, κ is the economy's profit share of spending, the expenditure-weighted average of industry margins. Wages account for the remaining share, so income equals the wage bill multiplied by $1/(1-\kappa)$. Income Y therefore depends on the profit composition of spending. It increases if the regulated industry has the higher margin ($Y^R > Y$ if $m_g > m_{-g}$), as spending shifts toward the higher-margin industry, and decreases in the opposite case. The results below hold regardless of the income response.

Lemma 1 (Industry expenditure never decreases)**.** *The log change in expenditure on industry g satisfies*

$$B \equiv \Delta \ln(S_g Y) \geq 0,$$

with equality only when $\sigma_0 = 1$.

Proof in Appendix B.

Even if income decreases in equilibrium, the increase in the industry's expenditure share more than offsets the decline.

Proposition 4 (Collective profitability)**.** *If* $\sigma_0 < 1$*, aggregate revenue and profit increase strictly in industry* g *and decrease strictly in every other industry. If* $\sigma_0 = 1$*, aggregate revenue and profit remain unchanged in every industry.*

Proof in Appendix B.

A cost-raising regulation is therefore collectively profitable for the regulated industry, with the gain coming from expenditure drawn away from other industries. The elasticity of incidence in Section 2.6 determines how unevenly the industry's higher total expenditure is distributed across firms.

4.3 Who wins

Before regulation, a firm with marginal cost ψ earns revenue $r(\psi) = \left(\mu_g \psi / P_g\right)^{1-\sigma_g} S_g Y$. After regulation, it earns $r^R(\psi) = \left(\mu_g \psi^R / P_g^R\right)^{1-\sigma_g} S_g^R Y^R$. Taking logs gives

$$\begin{aligned} \ln r(\psi) &= (1-\sigma_g)\left[\ln \mu_g + \ln \psi - \ln P_g\right] + \ln S_g + \ln Y\,, \\ \ln r^R(\psi) &= (1-\sigma_g)\left[\ln \mu_g + \ln \psi + \ln(1+\varepsilon(\psi)) - \ln P_g^R\right] + \ln S_g^R + \ln Y^R\,. \end{aligned}$$

Subtracting the pre-regulation equation from the post-regulation equation cancels the markup and the firm's baseline marginal cost:

$$\Delta \ln r(\psi) = (\sigma_g - 1)\left[\Delta \ln P_g - \ln(1+\varepsilon(\psi))\right] + B\,, \tag{24}$$

where $B = \Delta \ln S_g + \Delta \ln Y$ is the change in log industry expenditure from Lemma 1. The first term is the partial-equilibrium change in log revenue from Section 2: a firm gains when its own incidence lies below the rise in the industry's price index (15). General equilibrium adds the common term B to every firm's change in log revenue.

Theorem 2 (Who wins in general equilibrium)**.** *The change in log revenue,* $\Delta \ln r(\psi)$*, is strictly decreasing in the firm's baseline marginal cost* ψ*. A firm with cost* ψ *gains revenue and profit if and only if*

$$\ln\left(1+\varepsilon(\psi)\right) < \Delta \ln P_g + \frac{B}{\sigma_g - 1}.$$

The threshold on the right-hand side strictly exceeds $\ln(1+\varepsilon_0)$*, so the most productive firms always grow.*

Proof in Appendix B.

Because demand across industries is inelastic, the regulation draws expenditure into the industry (Lemma 1). This expenditure raises every firm's revenue by the same proportion and increases the partial-equilibrium threshold (Prop. 1) by $B/(\sigma_g - 1)$. General equilibrium therefore adds the same amount to each firm's log revenue change: winners gain more, losers lose less, and the set of winners expands. Changes in market share still follow the partial-equilibrium condition of Theorem 1.

Corollary 2 (Every firm gains under proportional incidence)**.** *Suppose* $\sigma_0 < 1$ *and let incidence be proportional,* $\varepsilon(\psi) = \varepsilon > 0$ *for every* ψ*. Then every firm in industry* g *gains revenue and profit, with*

$$\Delta \ln r(\psi) \;=\; B \;>\; 0 \qquad \textit{for every } \psi.$$

At $\sigma_0 = 1$*,* $B = 0$*. The regulation then becomes neutral as in partial equilibrium.*

Proof in Appendix B.

A proportional regulation moves no relative cost (Prop. 2). The inelastic outer nest still draws expenditure in, so every firm gains revenue and profit.

Remark 2 (Cobb–Douglas outer nest). When $\sigma_0 = 1$, the outer nest fixes industry expenditure shares, so $D = 1$ and $B = 0$. The regulation is zero-sum across firms in the regulated industry: winners' gains equal losers' losses. The condition for a firm to gain reduces to the partial-equilibrium condition (15). An inelastic outer nest turns this zero-sum reallocation into a positive-sum one for the industry.

Remark 3 (Elastic outer nest). For $\sigma_0 > 1$, expenditure leaves the industry as its price index increases. Section 4.6 analyzes this case.

4.4 The welfare cost

Welfare is real income, $W = Y/P_0$. Only the price index of industry g changes, so

$$\begin{aligned} \Delta \ln P_0 &= \frac{\ln D}{1-\sigma_0} \;\geq\; S_g \, \Delta \ln P_g, \\ \Delta \ln W &= \Delta \ln Y - \Delta \ln P_0, \end{aligned} \tag{25}$$

where $\Delta \ln Y = \ln\big[(1-\kappa)/(1-\kappa^R)\big]$ from (23). At $\sigma_0 = 1$, $\Delta \ln P_0 = S_g \, \Delta \ln P_g$.[21]

Proposition 5 (Welfare cost)**.** *With a common margin ($m_g = m_{-g}$), welfare decreases:*

$$\Delta \ln W = -\Delta \ln P_0 < 0.$$

With heterogeneous margins, welfare decreases to first order in $\Delta \ln P_g$:

$$\Delta \ln W^{(1)} = -\, S_g \, \Delta \ln P_g \left[1 - (1-\sigma_0)(1-S_g)\frac{m_g - m_{-g}}{1-\kappa}\right] < 0.$$

Proof in Appendix B.

This welfare cost reflects the resources absorbed by compliance. Compliance labor displaces production labor, raising the industry's price index and the cost of the same consumption basket. To first order, the loss is the category's expenditure share times its price-index rise, adjusted by the bracketed term. That term captures the income response and remains strictly positive. Any benefits of the regulation should be weighed against this cost.

These formulas require only aggregate and own-industry objects; the internal structure of the other industries does not enter.

4.5 Quantification

Assume inverse-Pareto productivity, $\Pr(\varphi < x) = (x/\varphi_{\max})^{\theta}$. Marginal cost, $\psi = w/\varphi$, then follows a Pareto distribution on $[\psi_{\min}, \infty)$, with $\Pr(\psi > y) = (\psi_{\min}/y)^{\theta}$ and $\psi_{\min} = w/\varphi_{\max}$. Under this parameterization, proportional regulation changes the scale parameter, whereas regressive regulation changes the shape parameter (Appendix E).

Assume a constant incidence elasticity β, as in Section 2.6, and let incidence vanish for the most productive firms: $\varepsilon_\omega \to \varepsilon_0 = 0$ as $\omega \to 0$. The incidence schedule is

$$1 + \varepsilon(\psi) = \left(\frac{\psi}{\psi_{\min}}\right)^{\beta} = \left(\frac{\varphi_{\max}}{\varphi}\right)^{\beta}. \tag{26}$$

Positive incidence for the most productive firms ($\varepsilon_0 > 0$) would add a proportional increase to this regressive schedule. This proportional component would increase the welfare cost, so the cost computed below is a lower bound.

Post-regulation marginal cost is $\psi^R = (1 + \varepsilon(\psi))\,\psi = \psi_{\min}\,(\psi/\psi_{\min})^{1+\beta}$. Its distribution

[21] The expression $D = (1 - S_g) \cdot 1 + S_g\,\nu$ is a weighted arithmetic mean of 1 and ν. The arithmetic–geometric mean inequality therefore gives $D \geq 1^{1-S_g}\,\nu^{S_g} = \nu^{S_g}$. Taking logs and dividing by $1 - \sigma_0 > 0$ yields $\Delta \ln P_0 \geq S_g \, \Delta \ln P_g$.

is

$$\begin{aligned}\Pr\big(\psi^R > y\big) &= \Pr\Big(\psi > \psi_{\min}\,(y/\psi_{\min})^{\frac{1}{1+\beta}}\Big) \\ &= \Big(\frac{\psi_{\min}}{y}\Big)^{\frac{\theta}{1+\beta}}.\end{aligned} \tag{27}$$

Regulation therefore shifts the shape parameter from θ to $\theta/(1+\beta)$, fattening the marginal-cost tail and increasing marginal-cost dispersion.

The price-index response in (20) also has a closed-form expression. Substituting the incidence schedule (26) and the Pareto density $dG(\psi) = \theta\,\psi_{\min}^{\theta}\,\psi^{-\theta-1}\,d\psi$ gives

$$\begin{aligned}\Big(\frac{P_g^R}{P_g}\Big)^{1-\sigma_g} &= \frac{\displaystyle\int_{\psi_{\min}}^{\infty} \psi^{1-\sigma_g}\big(\psi/\psi_{\min}\big)^{\beta(1-\sigma_g)}\,dG(\psi)}{\displaystyle\int_{\psi_{\min}}^{\infty} \psi^{1-\sigma_g}\,dG(\psi)} \\ &= \frac{\theta+\sigma_g-1}{\theta+(1+\beta)(\sigma_g-1)}.\end{aligned}$$

Taking logs yields

$$\Delta \ln P_g = \frac{1}{\sigma_g - 1}\ln\frac{\theta+(1+\beta)(\sigma_g-1)}{\theta+\sigma_g-1}. \tag{28}$$

This expression enters (21)–(25) directly. Given $(\theta, \sigma_g, \sigma_0, S_g)$, the estimated β therefore identifies the welfare cost. Appendix G reports standard errors for these objects and the results under alternative elasticities.

For β, I use the product-specific estimates from Section 3: the difference-in-differences estimate for products with flat pre-trends and the trend-adjusted post average for the others. I estimate the Pareto shape θ from each product's 2011 firm-size distribution. Under the Pareto parameterization, the log difference between the largest firm's baseline revenue and firm i's baseline revenue, $\ln(r_{\max}/r_i)$, follows an exponential distribution with rate $\theta/(\sigma_g - 1)$.[22] I estimate this rate as the reciprocal of the mean log revenue distance across firms, using the tail-index estimator of Hill (1975).

I set the within-industry elasticity to $\sigma_g = 3.9$, the median between-firm elasticity estimated by Hottman et al. (2016) using the same Nielsen data. The outer elasticity, σ_0, governs how spending on a product category responds to its own price index. I set $\sigma_0 = 0.5$, based on the category-level food price elasticities of 0.27 to 0.81 across the studies reviewed by Andreyeva et al. (2010).

The category's expenditure share, S_g, is its 2016 revenue scaled to the national market

[22] The log of a Pareto variable is exponentially distributed: $\Pr\big(\ln(\psi/\psi_{\min}) > t\big) = e^{-\theta t}$. Since $\ln(r_{\max}/r_i) = (\sigma_g - 1)\ln(\psi_i/\psi_{\min})$, the log revenue distance is exponential with rate $\theta/(\sigma_g - 1)$. The mean of an exponential variable is the reciprocal of its rate. Appendix F details the estimation.

and divided by personal consumption expenditure.[23]

A firm gains revenue and profit when its incidence lies below the cutoff in Theorem 2:

$$\ln(1+\varepsilon(\psi)) < \xi \equiv \Delta \ln P_g + \frac{B}{\sigma_g - 1}.$$

Under (26), this condition becomes $\beta \ln(\psi/\psi_{\min}) < \xi$, or $\psi/\psi_{\min} < e^{\xi/\beta}$. The Pareto CDF therefore gives a share of winners of $1 - e^{-\theta\xi/\beta}$.

I calculate the welfare cost under a common margin and express it in dollars as $-\Delta \ln W$ times 2016 personal consumption expenditure.

Product	β	θ	$\Delta \ln P_g$ (%)	B (%)	Firms gaining (%)	Welfare loss ($M/yr)
Nuts, oil-roasted	0.456	0.89	10.3	5.2	21	41
Nuts, dry-roasted	0.238	0.86	5.8	2.9	22	44
Dried fruit, true-dried	0.292	0.80	7.1	3.6	20	19
Peanut butter, creamy	0.103	0.74	2.7	1.4	21	26
Peanut butter, chunky	0.097	0.79	2.5	1.3	21	5
Chocolate, nut/PB	0.039	0.57	1.1	0.6	17	27
Granola bars, w/ inclusions	0.348	0.74	8.4	4.2	19	56
Granola cereal	0.256	1.26	5.7	2.8	28	12
Cereal	0.064	0.54	1.8	0.9	16	105
Seasoning	0.049	0.64	1.4	0.7	19	15
Crackers, flavored	0.027	0.57	0.8	0.4	17	11

Table 5: Quantification by treated product. All calculations use $\sigma_g = 3.9$ and $\sigma_0 = 0.5$. The incidence elasticity β is estimated in Section 3, and θ is the Pareto shape parameter fitted to the 2011 firm-size distribution. The log price-index change $\Delta \ln P_g$ is given by (28), and B is the log change in category expenditure (Lemma 1, under the common margin $\Delta \ln Y = 0$); both are reported in percent. *Firms gaining* is the share of firms, by count, below the cutoff in Theorem 2, expressed in percent. Annual welfare costs are calculated under a common margin as $-\Delta \ln W$ times 2016 personal consumption expenditure and reported in millions of dollars.

Table 5 reports the results. In oil-roasted nuts, the estimated incidence elasticity of 0.456 fattens the marginal-cost tail, reducing the shape parameter from $\theta = 0.89$ to 0.61. The category's price index increases by 10%, and expenditure on the category increases by 5%. The winners are the most productive fifth of firms, and the annual welfare cost is $41

[23] I first divide sampled revenue by the share of category revenue across all Nielsen stores accounted for by stores in the balanced sample (0.83 to 0.88). I then divide by the panel's coverage of food-at-home sales, about 42%, calculated from Nielsen's channel coverage weighted by USDA's outlet shares. The denominator is 2016 personal consumption expenditure, $12.75 trillion (BEA, NIPA Table 2.3.5). The resulting category shares range from 0.002% to 0.05% of expenditure.

million.

In every category, between a sixth and a third of firms gain. Across the eleven treated products, annual welfare costs total $362 million, with cereal, the largest category, accounting for more than a quarter of the total.

4.6 The elastic outer nest

When $\sigma_0 > 1$, industries are gross substitutes: expenditure leaves industry g as its price index increases. The sign of the expenditure response in Lemma 1 therefore reverses.

Proposition 6 (Industry expenditure decreases)**.** *For $\sigma_0 > 1$, expenditure on industry g decreases strictly: $B < 0$.*

Proof in Appendix B.

In (24), every firm's change in log revenue now falls short of its partial-equilibrium value by the common amount $|B|$. Winners gain less, losers lose more, and the set of winners shrinks. General equilibrium therefore mutes the mechanism.

As in Lemma 1, this result holds regardless of the income response. Income increases when expenditure shifts to industries with higher margins than industry g's and decreases when it shifts to industries with lower margins. In either case, the income response never outweighs the decline in industry g's expenditure share.

The sign of B determines how general equilibrium changes the mechanism: $\sigma_0 < 1$ amplifies it, $\sigma_0 = 1$ leaves it at its partial-equilibrium value (Remark 2), and $\sigma_0 > 1$ mutes it.

The most productive firms can still grow.

Theorem 3 (Winners under an elastic outer nest)**.** *Let $\sigma_0 > 1$ and $\sigma_g > \sigma_0$. Suppose incidence is regressive, with $\varepsilon(\psi_{\min}) = 0$. Under a common margin across industries, the most productive firm gains revenue and profit:*

$$\Delta \ln r(\psi_{\min}) \;>\; (\sigma_g - \sigma_0)\, \Delta \ln P_g \;>\; 0.$$

Because $\Delta \ln r(\psi)$ is strictly decreasing in ψ, the winning firms form a top segment of the productivity distribution. With heterogeneous margins, the same conclusions hold if $\sigma_h \geq 2$ for every industry h.

Proof in Appendix B.

With expenditure leaving the industry, regressive incidence no longer guarantees a gain for the most productive firms. It is sufficient that the within-industry elasticity exceeds the

across-industry one, so that enough demand flows from the least to the most productive firms, and that the incidence of the most productive vanishes.

5 Welfare in Homogeneous Goods

This final exercise asks what the raising-rivals'-costs tradition would conclude about the welfare effects of FSMA. I use the simplest model in that tradition: a homogeneous good supplied by a low-cost leader and n identical higher-cost followers. The firms compete à la Cournot under linear demand $p = a - bQ$. The regulation raises the followers' marginal cost c_f and spares the leader's cost c_L. Welfare is total surplus: the sum of consumer and producer surplus.

Salop and Scheffman (1987, Proposition 5) established that the welfare effect is ambiguous because the regulation also reallocates output from high-cost to low-cost firms, causing a gain in efficiency. The sign is determined by a closed-form threshold for the leader's market share. The leader crosses this threshold in four of the categories studied in Section 3.

Appendix M repeats the analysis in the dominant-firm structure of Salop and Scheffman (1987). Appendix N calibrates a differentiated demand system that reproduces the homogeneous model's price, shares, and markups.

5.1 The leader's share measures the efficiency gap

In the Cournot equilibrium each firm's first-order condition equates its price–cost difference to b times its own quantity, $p - c_i = b\,q_i$. Dividing the leader's condition by a follower's,

$$\frac{q_L}{q_f} \;=\; \frac{p - c_L}{p - c_f}\,,$$

so the leader's share is determined by its cost relative to the followers'. Summing the $n+1$ first-order conditions gives $b\,Q = (n+1)\,p - c_L - n\,c_f$, and substituting demand, $b\,Q = a - p$, solves for the price:

$$p \;=\; \frac{a + c_L + n\,c_f}{n+2}\,.$$

Each quantity is then its price–cost difference over b, and taking ratios,

$$\frac{q_L}{q_f} \;=\; \frac{(a - c_L) + n\,(c_f - c_L)}{(a - c_L) - 2\,(c_f - c_L)}\,, \qquad s_L \;=\; \frac{(a - c_L) + n\,(c_f - c_L)}{(n+1)(a - c_L) - n\,(c_f - c_L)}\,,$$

where s_L is the leader's share of output and of revenue. Both expressions increase in the cost gap $c_f - c_L$ relative to $a - c_L$. Given the number of followers, the leader's share is a

one-to-one function of that efficiency gap. This is what lets us determine the sign of the welfare change knowing only the leader's share and the number of firms.

5.2 Why surplus can increase

Total surplus is the area under the demand curve minus production costs. A small increase in the followers' cost changes it in two ways. The followers pay the higher cost on every unit they produce, the direct burden. But output reallocates from the least to the most efficient firm. A unit change in output from firm i moves surplus by $p - c_i$, what consumers pay for the marginal unit net of what it costs to make. Together,

$$dTS \;=\; -\,n\,q_f\,dc_f \;+\; \sum_i (p - c_i)\,dq_i\,,$$

the single-industry form of the wedge formula of Harberger (1971) and Baqaee and Farhi (2020). The leader expands and the followers contract. A unit of output that moves from a follower to the leader is valued at the difference between the two price–cost differences, $(p-c_L)-(p-c_f) = c_f - c_L$. Every unit reallocated is registered as a resource saving equal to the cost gap. Against this "credit" stand the direct cost of the burden and the contraction of total output, both proportional to the followers' output. A large enough gap makes the credit on the reallocated units outweigh both, and the regulation raises total surplus.

5.3 The threshold

Proposition 7 (When the homogeneous model reports a gain)**.** *The marginal regulation ($dc_f > 0$, leader spared) changes total surplus by*

$$\frac{dTS}{dc_f} \;=\; \frac{n}{n+2}\left[\,q_L - (n+4)\,q_f\,\right],$$

so total surplus increases if and only if

$$\frac{c_f - c_L}{a - c_L} \;>\; \frac{n+3}{3n+8} \qquad \textit{equivalently} \qquad s_L \;>\; s_L^*(n) \;\equiv\; \frac{n+4}{2(n+2)} \;\in\; \left(\tfrac{1}{2},\,\tfrac{5}{6}\right].$$

Proof in Appendix M. The threshold depends on the primitives only through (s_L, n), declines as the number of firms grows, converging to one half.

Section 4 computed the welfare change from the regulation in the differentiated economy, finding a loss in every product. Seen through the lens of the homogeneous model, FSMA resulted in welfare gains in four of those product categories. At the firm counts of Table 2, the

threshold goes from 0.53 to 0.58, and the leader sits above it in dry-roasted nuts (Wonderful, 0.64 against 0.53), flavored crackers (Mondelez, 0.63 against 0.54), seasoning (McCormick, 0.56 against 0.53), and creamy peanut butter (Smucker's Jif, 0.57, at its threshold). A researcher who took the homogeneous model to these differentiated goods would instead report that the regulation raised welfare in each of them.

6 Conclusion

Firms can benefit from a regulation that raises production costs, including their own. When incidence is asymmetric, every firm's price rises, but the prices of the least burdened firms rise the least. These firms become relatively cheaper, attract demand, and gain revenue, profit, and market share. The mechanism does not require exemptions, barriers to entry, or the forced exit of rivals. It is distinct from the output reallocation of *raising rivals' costs*, which it amplifies.

I focus on regulations with *regressive incidence*, whose incidence decreases with productivity. This occurs when the most productive firms already run the required systems, use less of the regulated input, or pay above the proposed floor. In a differentiated industry with homothetic single-aggregator demand, a firm gains if and only if the price aggregator increases proportionally more than its own marginal cost. Under regressive incidence, the most productive firms grow and the least productive shrink.

Under CES, log revenue is linear in log productivity. The slope is the *productivity–size gradient*, and its change when regulation takes effect estimates the elasticity of incidence. Regulations with proportional incidence and those that raise fixed costs leave the gradient unchanged. Progressive incidence flattens it, and only regressive incidence steepens it.

I find evidence of this mechanism in the Food Safety Modernization Act. In Nielsen scanner data around the September 2016 compliance date, the productivity–size gradient steepens in product categories where the *Salmonella*-control obligation is the firm's own. It does not move in closely related products whose terminal kill step or supplier verification allows them to avoid that obligation. The reallocations are economically large: in oil-roasted nuts, the sales ratio between two firms rose from 10:1 before regulation to nearly 30:1 by the end of the post period. The largest firms in these categories and their trade associations supported and lobbied for the statute, while the very smallest producers fought for an exemption.

In a nested CES economy with an inelastic outer nest, general equilibrium amplifies the mechanism: the industry's higher price index draws expenditure from other industries. With Pareto marginal costs, the estimated coefficients map into the distribution's shape

parameter. The estimates imply category price indices up to ten percent higher and gains for only the top sixth to third of firms. Across the treated categories, the implied welfare cost is at least $362 million a year. The homogeneous-good model of the raising-rivals'-costs tradition would report welfare gains in four of those categories because it treats reallocation toward the lower-cost leader as an efficiency gain.

This paper takes regulations as given and asks who wins. The natural next step is to study how regulations are created and how firms lobby over the shape of the incidence schedule.

The results change how a cost-raising regulation should be evaluated. The relevant question is not only how much a regulation costs in aggregate or whether it formally applies to every firm, but how its incidence varies across the affected industry. A regulation can be universal in law and sharply asymmetric in effect. Its incidence schedule is itself an object of political economy and should be measured alongside the regulation's direct benefits. By changing relative marginal costs and hence relative prices, regulation reallocates demand, raising costs and concentrating the market in the same move. It picks winners and losers.

A The Literature in Cost-Function Space

A.1 A common structure

The models this paper builds on and those in the literature on cost-increasing regulation differ in market structure and the type of goods. Their market structures include Cournot competition, a dominant firm facing a fringe, and monopolistic competition; goods may be homogeneous or differentiated. They all share one and the same description of the firm. Firm i solves

$$\max_{q_i} \; p_i\, q_i - C_i(q_i), \qquad C_i(q_i) = c_i\, q_i + F. \tag{29}$$

Here, c_i is a constant marginal cost that may differ across firms. The fixed cost F is common across firms when present and is often absent.

The raising-rivals'-costs tradition uses heterogeneous c_i and typically sets $F = 0$ (Williamson, 1968; Salop and Scheffman, 1983, 1987). Melitz-type models write $c_i = w/\varphi_i$ for a productivity draw φ_i and use the common fixed cost F to generate exit (Melitz, 2003; Macedoni and Weinberger, 2026).

In this formulation, a regulation deforms the distribution of marginal costs c_i or shifts the common fixed cost F. The analysis is a comparative static with respect to that change. This paper studies marginal-cost deformations in differentiated goods. Before organizing the literature, I establish when it is legitimate to posit the cost function (29) directly, as this paper and every related paper do. Section A.2 gives the exact condition. Section A.4 then organizes the related work by cost deformation and type of goods.

A.2 Isolation: when cost functions are the primitives

Firm i has a production technology $q_i = f_i(x_i)$ that maps an input vector x_i into output and faces factor prices r.

Definition 2 (Isolated cost minimization)**.** *Firm i's cost-minimization problem is*

$$C_i(q_i) \equiv \min_{x_i} \big\{\, r \cdot x_i : f_i(x_i) \geq q_i \,\big\}.$$

The problem is isolated *if firm i takes the factor prices r as given and no individual firm's choices affect r.*

Isolation is compatible with both partial and general equilibrium. Factor prices r can be determined in equilibrium through factor-market clearing. Isolation requires only that no individual firm can influence those prices.

Proposition A.1 (Reduction to cost-function space)**.** *Suppose cost minimization is isolated for every firm. Firm i's problem is equivalent to the following two-stage problem. First, compute $C_i(q_i)$ as a function of own output and the given factor prices alone. Second, choose a point on the demand curve to maximize $p_i q_i - C_i(q_i)$. Industry equilibrium is characterized entirely by the cost functions $\{C_i\}$ and demand. Every primitive source of firm heterogeneity enters only through $\{C_i\}$.*

The proof is in Appendix B.

Cost minimization requires no knowledge of rivals. Because no individual firm's choices affect r, the cost function depends only on own output and the given factor prices. As in any oligopoly, rivals enter only at the second stage, through firm i's residual demand curve. Under monopolistic competition, they enter through the aggregator that each firm takes as given. Duality ensures that the first stage recovers all information relevant to production.

Remark A.1 (The related work is already here). Melitz-type models take the productivity distribution as given and write marginal cost as $\psi = w/\varphi$. This formulation combines constant returns to scale, a common input cost w, and marginal-cost heterogeneity determined by productivity φ. Models in the raising-rivals'-costs tradition posit constant marginal costs directly. Under isolation, the production function is redundant, so starting from the cost function entails no loss of generality.

A.3 Affine form

The preceding conditions allow us to work directly with the cost functions $\{C_i\}$. The affine form (29) requires two further restrictions: constant returns in variable inputs and a common fixed overhead F. Under constant returns, producing q units costs q times the minimized cost of one unit, so variable cost is linear in output. Hence,

$$C_i(q_i) = c_i\, q_i + F.$$

When technologies differ only in productivity, as in the classic Melitz model, marginal cost is $c_i = e(r)/\varphi_i$, where $e(r)$ is the unit cost of the common input bundle. For a regulation that raises input prices r to have asymmetric incidence, the input bundle must be firm-specific, so that $c_i = e_i(r)/\varphi_i$. Incidence on firm i is then its expenditure share on the regulated factor.

Appendix K develops two settings in which regulations that raise factor prices generate regressive incidence. The first is a uniform labor tax in an industry where capital intensity increases with productivity. The second is a minimum wage regulation in a setting where firm-specific wages increase with productivity, as in Helpman et al. (2010).

Next, I organize the relevant literature.

A.4 Four ways to create winners

The question traces back to classic capture theory (Stigler, 1971; Peltzman, 1976), in which regulation kept rivals out. By controlling entry, captured regulation restricted supply and generated rents for incumbents. The next generation shifted from entry barriers to production costs: firms could gain from regulation when it burdened their rivals more. Every model in this second generation falls into one of the four cells in Table 6.

Homogeneous goods with marginal-cost incidence form the largest cell, the *raising rivals' costs* tradition. The corresponding fixed-cost cell is nearly empty. For differentiated goods, fixed-cost incidence belongs to the *selection* tradition. The few papers studying marginal-cost incidence in differentiated goods find neutrality. This paper traces their neutrality result to the particular deformation they study: proportional incidence in a homothetic environment (Prop. 2).

	Homogeneous goods	**Differentiated goods**
Marginal cost	Raising rivals' costs (Salop and Scheffman, 1983; Fowlie et al., 2016)	Regressive incidence (*this paper*); neutrality under proportional incidence (Macedoni and Weinberger, 2026; Anouliès, 2017)
Fixed cost	Exclusion (Hviid and Olczak, 2016)	Selection (Macedoni and Weinberger, 2026; Rebeyrol, 2023)

Table 6: How a cost-raising regulation creates within-industry winners. Columns distinguish homogeneous from differentiated goods, and rows distinguish regulations that raise marginal costs from those that raise fixed costs. The cells identify mechanisms and representative studies.

Homogeneous goods, marginal cost. This is the oldest and largest cell. Williamson (1968) models the Pennington case, in which a union wage burdens small, labor-intensive mines more than the large mines that negotiated for it. Salop and Scheffman (1983, 1987) give the tradition its name and canonical structure: a dominant firm profits from raising the marginal cost of a price-taking fringe. Michaelis (1994) extends this logic to Cournot competition, where the low-cost firm gains market share from regulation that burdens every firm asymmetrically.

Modern environmental applications build on the same mechanism. Fowlie et al. (2016) incorporate a carbon price into cement producers' marginal costs in a dynamic Cournot industry, where cleaner firms gain output from dirtier rivals. Fowlie (2009) finds the same

reallocation in a static oligopoly under incomplete regulation. Baccianti and Schenker (2022) bring incomplete coverage into general equilibrium, where spared firms gain through higher markups.

This cell also includes models without strategic interaction. In electricity markets, a carbon price changes the merit order, and the single market price increases with the marginal unit's cost. Clean inframarginal generators collect the difference as Ricardian rents. Fabra and Reguant (2014) measure this effect and find near-complete pass-through.

In Muñoz-García and Akhundjanov (2016), incidence runs in the opposite direction: regulation favors the least efficient firm.

Grey (2018) adds a second stage with lobbying: a firm first adopts a clean technology and then lobbies for the emissions tax its rival will pay. In Cai and Li (2020), a sufficiently large clean firm lobbies for higher pollution taxes. Kennard (2020) documents the same behavior in the record of climate regulation.

Empirical applications of this tradition include the 1833 Factory Act, OSHA, the Clean Air Act, and modern cap-and-trade (Marvel, 1977; Oster, 1982; Maloney and McCormick, 1982; Landes, 1980; Pashigian, 1984; Bartel and Thomas, 1985, 1987; Curtis and Lee, 2018).

Homogeneous goods, fixed cost. This cell contains few papers. Hviid and Olczak (2016) show that raising rivals' fixed costs can profitably force exit even when the strategist's own fixed cost increases by as much or more. Fixed costs affect the participation constraint rather than the price. Becker and Henderson (2000) empirically study air-quality regulation whose fixed compliance costs fall on new plants while incumbents are grandfathered in.

Differentiated goods, fixed cost. This is the selection tradition. In Melitz (2003), a common fixed cost and a productivity distribution generate an exit cutoff. Raising the fixed cost raises the cutoff: the least productive firms exit, and survivors absorb their demand. Macedoni and Weinberger (2026) introduce regulation into this framework and derive its political implications. Their fixed-cost product standard benefits large firms, which lobby for it. In Rebeyrol (2023), a regulation that raises every firm's fixed cost by the same amount benefits the survivors that inherit exiting firms' demand. Abel-Koch (2013) derives the politics of this mechanism in trade, where the largest firms secure such regulations through lobbying. Andersen (2018) measures the welfare cost of regulation in this environment and finds that the direct compliance cost understates the welfare loss.

Differentiated goods, marginal cost. This cell contains three neutrality results. Macedoni and Weinberger (2026) prove that a regulation acting on marginal costs benefits no firm of any size, so no firm lobbies for it. Anouliès (2017) finds that a cap-and-trade scheme is neutral for revenues and composition. Both results assume that regulation raises marginal

costs by a common proportional factor. The same neutrality holds, unremarked, in Forslid et al. (2018). After firms re-optimize abatement, the emissions tax multiplies every marginal cost by a common factor, making the regulation's marginal-cost channel neutral.

Section 2 shows that this neutrality is a property of proportional incidence, not of marginal-cost regulation itself. What this cell lacked was the deformation studied in the raising-rivals'-costs tradition: regressive incidence over marginal costs. Demirer et al. (2026) find regressive incidence in the European Union's General Data Protection Regulation: a marginal-cost wedge that decreases with firm size.

This paper embeds raising rivals' costs in differentiated goods and brings the mechanism to general equilibrium.

B Proofs

Theorem 1 (Who wins). *Under a deformation $\psi \to \psi^R$, firm ω strictly gains revenue, profit, and market share if and only if its relative marginal cost decreases:*

$$\frac{\psi_\omega^R}{P(\boldsymbol{\mu}\boldsymbol{\psi}^R)} < \frac{\psi_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi})}.$$

All three outcomes strictly decrease if the inequality is reversed and remain unchanged under equality.

Proof of Theorem 1. Section 2.1 establishes that firm ω's equilibrium profit, revenue, and market share are strictly decreasing in its relative marginal cost ψ_ω/P. For profit, this follows from the envelope condition (6). Revenue $Y s_\omega(z_\omega^*)$ and market share $s_\omega(z_\omega^*)$ decrease because the optimal relative price increases with relative cost while the share function decreases. Each outcome therefore improves if and only if regulation lowers the relative-cost ratio.

Define $\rho \equiv P(\boldsymbol{\mu}\boldsymbol{\psi}^R)/P(\boldsymbol{\mu}\boldsymbol{\psi})$. Rearranging gives

$$\frac{\psi_\omega^R}{P(\boldsymbol{\mu}\boldsymbol{\psi}^R)} < \frac{\psi_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi})} \quad \Longleftrightarrow \quad \frac{\psi_\omega^R}{\psi_\omega} < \frac{P(\boldsymbol{\mu}\boldsymbol{\psi}^R)}{P(\boldsymbol{\mu}\boldsymbol{\psi})} = \rho.$$

The loss and equality cases follow by reversing the inequality or replacing it with equality. In the equality case, firm ω's relative cost is unchanged, so its relative price and market share are unchanged. With industry expenditure fixed, its revenue $s_\omega Y$ and profit are also unchanged. □

Lemma B.1 (The equilibrium aggregator in cost space)**.** *Write $P^*(\psi)$ for the equilibrium aggregator under the cost profile ψ, so that $P^*(\psi) = P(\boldsymbol{\mu}\boldsymbol{\psi})$ at the equilibrium markup profile.*

(i) If $\psi'_\omega \geq \psi_\omega$ for almost every ω, with strict inequality on a set of positive measure, then $P^*(\psi') > P^*(\psi)$.

(ii) $P^*(\lambda\psi) = \lambda\, P^*(\psi)$ for every $\lambda > 0$.

Proof. Profit (3) has strictly increasing differences in the relative price z and the relative cost $c \equiv \psi_\omega/P$: the cross-partial is $s_\omega(z)/z^2 - s'_\omega(z)/z > 0$ under gross substitutes. The optimal relative price $z^*_\omega(c)$ is therefore increasing in c, and strictly so because the first-order condition (4) maps each z^*_ω to a single c.[24] Markups adjust freely along this map.

In equilibrium, adding-up (2) reads

$$\int_0^1 s_\omega\big(z^*_\omega(\psi_\omega/P)\big)\, d\omega \;=\; 1.$$

The integrand is strictly decreasing in ψ_ω and strictly increasing in P, since $s'_\omega < 0$.

(i) Evaluate the integral at ψ' and at the original aggregator $P^*(\psi)$. Every relative cost weakly increases, and strictly on a set of positive measure, so the integral is strictly below one. Restoring adding-up requires a strictly larger aggregator.

(ii) Scaling every cost and the aggregator by λ leaves every relative cost, hence every z^*_ω, unchanged, so adding-up still holds. Uniqueness of the aggregator gives the result. □

Proposition B.1 (Exact characterization of gains at the top)**.** *Let $\rho \equiv P(\boldsymbol{\mu}\boldsymbol{\psi}^R)/P(\boldsymbol{\mu}\boldsymbol{\psi})$. Every firm in some neighborhood $(0, \delta)$ of the top of the productivity distribution gains if and only if $\psi^R_\omega/\psi_\omega < \rho$ for all ω in that neighborhood. In limit form, a sufficient condition is*

$$\limsup_{\omega \downarrow 0} \frac{\psi^R_\omega}{\psi_\omega} < \rho,$$

and a necessary condition is

$$\limsup_{\omega \downarrow 0} \frac{\psi^R_\omega}{\psi_\omega} \leq \rho.$$

Proof. By Theorem 1, firm ω gains if and only if $\psi^R_\omega/\psi_\omega < \rho$. If the limsup is strictly below ρ, this inequality holds throughout a neighborhood of the top, proving sufficiency. If the limsup exceeds ρ, firms with $\psi^R_\omega/\psi_\omega > \rho$ lie arbitrarily close to the top. No neighborhood consisting entirely of gainers can then exist, proving necessity of the weak inequality. Equality is consistent with gains when the ratio approaches ρ from below. □

Conditions (i)–(ii) of Corollary C.2 give an interpretable special case. Vanishing burden makes the limsup equal to one, and any increase in the aggregator puts ρ above one. The

[24] Assumption S of Matsuyama and Ushchev (2020), maintained throughout, makes this solution unique, so $z^*_\omega(c)$ is a function rather than a correspondence. CES satisfies it.

exact condition requires incidence along the top of the productivity distribution to fall short of the economy-wide price response. It therefore allows "vanishing" to be relaxed to "sufficiently small relative to the aggregate rise."

Proposition 1 (Regressive incidence). *Take a cost-increasing regulation and write its incidence as* $1+\varepsilon_\omega \equiv \psi^R_\omega/\psi_\omega$. *Suppose incidence is regressive:* $\varepsilon_\omega \le \varepsilon_{\omega'}$ *whenever* $\omega < \omega'$, *with strict inequality for some pair. Then there exist* $0 < \bar\omega \le \bar\omega' < 1$ *such that every firm in* $(0, \bar\omega)$ *strictly gains revenue, profit, and market share, and every firm in* $(\bar\omega', 1)$ *strictly loses.*

Proof of Proposition 1. Define the endpoint incidences by

$$1+\varepsilon_0 \equiv \lim_{\omega\downarrow 0}(1+\varepsilon_\omega),$$
$$1+\varepsilon_1 \equiv \lim_{\omega\uparrow 1}(1+\varepsilon_\omega) \in (1+\varepsilon_0, \infty].$$

Both limits exist because incidence is monotone. The strict inequality for some pair implies $\varepsilon_1 > \varepsilon_0$.

The winning segment. The cost profile satisfies $\psi^R_\omega = (1+\varepsilon_\omega)\psi_\omega \ge (1+\varepsilon_0)\psi_\omega$ almost everywhere, with strict inequality on a set of positive measure. Lemma B.1 therefore gives

$$P(\boldsymbol{\mu\psi}^R) = P^*(\psi^R) > P^*\big((1+\varepsilon_0)\,\psi\big) = (1+\varepsilon_0)\,P(\boldsymbol{\mu\psi}).$$

Thus $\rho > 1+\varepsilon_0$. Since $\psi^R_\omega/\psi_\omega \to 1+\varepsilon_0$ as $\omega \downarrow 0$,

$$\limsup_{\omega\downarrow 0} \frac{\psi^R_\omega}{\psi_\omega} = 1+\varepsilon_0 < \rho.$$

Proposition B.1 therefore gives a segment of winners at the top. By monotonicity of incidence, the set $\{\omega : 1+\varepsilon_\omega < \rho\}$ is an interval adjacent to 0, namely $(0, \bar\omega)$.

The losing segment. If $\varepsilon_1 = \infty$, then $\psi^R_\omega/\psi_\omega = 1+\varepsilon_\omega > \rho$ for every ω sufficiently close to one.

If $\varepsilon_1 < \infty$, monotonicity and the strict inequality for some pair imply $\psi^R_\omega \le (1+\varepsilon_1)\psi_\omega$ almost everywhere, with strict inequality on a set of positive measure. Lemma B.1 gives

$$P(\boldsymbol{\mu\psi}^R) < (1+\varepsilon_1)\,P(\boldsymbol{\mu\psi}),$$

so $\rho < 1+\varepsilon_1$. Since $1+\varepsilon_\omega \to 1+\varepsilon_1$ as $\omega \uparrow 1$, $\psi^R_\omega/\psi_\omega > \rho$ for every ω sufficiently close to one.

In either case, Theorem 1 gives strict losses near one. By monotonicity of incidence, the set $\{\omega : 1+\varepsilon_\omega > \rho\}$ is an interval adjacent to 1, namely $(\bar\omega', 1)$. □

Corollary 1 (Exit amplifies). *Suppose, in addition to the conditions of Proposition 1, that firms must pay a fixed operating cost $F > 0$ to remain active and that a positive measure of firms exits under regulation. Then the surviving winners in $(0, \bar{\omega})$ gain strictly more than in the no-exit case.*

Proof of Corollary 1. The unit continuum $(0, 1)$ is the set of firms active before the regulation, so the same regulation without exit has active set $(0, 1)$. Let $\Omega^R \subset (0, 1)$ be the post-regulation active set; by hypothesis, $(0, 1) \setminus \Omega^R$ has positive measure. As in the proof of Lemma B.1, write each surviving firm's optimized share as $s_\omega\big(z^*_\omega(\psi^R_\omega/P)\big)$, a strictly decreasing function of its relative cost in which the markup adjusts along z^*_ω. Let $P^R(\cdot)$ denote the aggregator defined by adding-up over Ω^R:

$$\int_{\Omega^R} s_\omega\big(z^*_\omega(\psi^R_\omega/P^R)\big)\, d\omega = 1.$$

Evaluate this integral at the no-exit aggregator, $P^R = P(\boldsymbol{\mu}\boldsymbol{\psi}^R) = P^*(\psi^R)$. The integral falls short of one because the integrand is unchanged but the domain has shrunk. Since each optimized share is strictly increasing in the aggregator, restoring adding-up requires

$$P^R(\boldsymbol{\mu}\boldsymbol{\psi}^R) > P(\boldsymbol{\mu}\boldsymbol{\psi}^R).$$

Hence, for every surviving $\omega \in (0, \bar{\omega}) \cap \Omega^R$,

$$\frac{\psi^R_\omega}{P^R(\boldsymbol{\mu}\boldsymbol{\psi}^R)} < \frac{\psi^R_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi}^R)} < \frac{\psi_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi})}.$$

The second inequality follows from whichever of Corollaries C.2 and C.1, or Proposition 1, applies, with $(0, \bar{\omega})$ denoting that result's winning set.

The argument requires only that the active set shrinks. The exit condition, variable profit falling below the operating fixed cost F for the most burdened firms, matters only through the resulting set Ω^R. Variable profit is strictly decreasing in relative cost. Since regulation leaves F unchanged, profit net of F increases strictly more than in the no-exit case. □

Proposition 2 (Proportional increases are neutral). *Let $\psi^R_\omega = (1+\varepsilon)\psi_\omega$ for all ω and some $\varepsilon > 0$. Then every firm's relative marginal cost is unchanged, as are its revenue, profit, and market share.*

Proof of Proposition 2. With $\psi^R_\omega = (1+\varepsilon)\psi_\omega$ for every ω, the scaled price profile is again an equilibrium. Scaling every price and the aggregator by $1+\varepsilon$ leaves every relative price and relative cost ψ_ω/P unchanged. Each firm's optimum in relative terms is therefore unchanged.

By linear homogeneity of the aggregator,

$$\rho = \frac{P\big((1+\varepsilon)\,\boldsymbol{\mu\psi}\big)}{P(\boldsymbol{\mu\psi})} = 1+\varepsilon = \frac{\psi_\omega^R}{\psi_\omega} \qquad \text{for every } \omega.$$

This is the equality case of Theorem 1, so market shares are unchanged. With industry expenditure fixed, revenue and profit levels are also unchanged. □

Proposition 3 (Progressive incidence). *Take a cost-increasing regulation and write its incidence as* $1+\varepsilon_\omega \equiv \psi_\omega^R/\psi_\omega$. *Suppose incidence is progressive:* $\varepsilon_\omega \geq \varepsilon_{\omega'}$ *whenever* $\omega < \omega'$, *with strict inequality for some pair. Then there exist* $0 < \bar{\omega} \leq \bar{\omega}' < 1$ *such that every firm in* $(0, \bar{\omega})$ *strictly loses revenue, profit, and market share, and every firm in* $(\bar{\omega}', 1)$ *strictly gains.*

Proof of Proposition 3. Define the endpoint incidences by

$$1+\varepsilon_1 \equiv \lim_{\omega\uparrow 1}(1+\varepsilon_\omega),$$
$$1+\varepsilon_0 \equiv \lim_{\omega\downarrow 0}(1+\varepsilon_\omega) \in (1+\varepsilon_1, \infty].$$

Both limits exist because incidence is monotone. The strict inequality for some pair implies $\varepsilon_0 > \varepsilon_1$.

The winning segment. The cost profile satisfies $\psi_\omega^R = (1+\varepsilon_\omega)\psi_\omega \geq (1+\varepsilon_1)\psi_\omega$ almost everywhere, with strict inequality on a set of positive measure. Lemma B.1 therefore gives

$$P(\boldsymbol{\mu\psi}^R) = P^*(\psi^R) > P^*\big((1+\varepsilon_1)\,\psi\big) = (1+\varepsilon_1)\,P(\boldsymbol{\mu\psi}).$$

Thus $\rho > 1+\varepsilon_1$. Since $\psi_\omega^R/\psi_\omega \to 1+\varepsilon_1$ as $\omega \uparrow 1$, $\psi_\omega^R/\psi_\omega < \rho$ for every ω sufficiently close to one. Theorem 1 therefore gives strict gains near one. By monotonicity of incidence, the set $\{\omega : 1+\varepsilon_\omega < \rho\}$ is an interval adjacent to 1, namely $(\bar{\omega}', 1)$.

The losing segment. If $\varepsilon_0 = \infty$, then $\psi_\omega^R/\psi_\omega = 1+\varepsilon_\omega > \rho$ for every ω sufficiently close to zero.

If $\varepsilon_0 < \infty$, monotonicity and the strict inequality for some pair imply $\psi_\omega^R \leq (1+\varepsilon_0)\psi_\omega$ almost everywhere, with strict inequality on a set of positive measure. Lemma B.1 gives

$$P(\boldsymbol{\mu\psi}^R) < (1+\varepsilon_0)\,P(\boldsymbol{\mu\psi}),$$

so $\rho < 1+\varepsilon_0$. Since $1+\varepsilon_\omega \to 1+\varepsilon_0$ as $\omega \downarrow 0$, $\psi_\omega^R/\psi_\omega > \rho$ for every ω sufficiently close to zero.

In either case, Theorem 1 gives strict losses near zero. By monotonicity of incidence, the set $\{\omega : 1 + \varepsilon_\omega > \rho\}$ is an interval adjacent to 0, namely $(0, \bar{\omega})$. □

Lemma 1 (Industry expenditure never decreases). $B \equiv \Delta \ln(S_g Y) \geq 0$, *with equality only at* $\sigma_0 = 1$.

Proof of Lemma 1. Write $E_g \equiv S_g Y$ for expenditure on industry g, so $B = \Delta \ln S_g + \Delta \ln Y$. Equation (22) gives $S_g^R = S_g \nu / D$. Since $D - S_g \nu = 1 - S_g$, it also gives $1 - S_g^R = (1 - S_g)/D$. Substituting into $\kappa^R = S_g^R m_g + (1 - S_g^R) m_{-g}$ yields

$$D\,\kappa^R = S_g\,\nu\,m_g + (1 - S_g)\,m_{-g}.$$

From (23), $\Delta \ln Y = \ln\big[(1-\kappa)/(1-\kappa^R)\big]$. Hence,

$$B = \ln\big[\nu(1-\kappa)\big] - \ln\big[D(1-\kappa^R)\big].$$

The sign of B is therefore the sign of

$$\begin{aligned}\nu(1-\kappa) - D(1-\kappa^R) &= \nu - D - \nu\big[S_g m_g + (1-S_g)m_{-g}\big] \\ &\quad + S_g \nu\, m_g + (1-S_g)m_{-g} \\ &= (\nu - D) - (1-S_g)(\nu-1)m_{-g} \\ &= (1-S_g)(\nu-1)(1-m_{-g}).\end{aligned}$$

The last equality uses $\nu - D = (1 - S_g)(\nu - 1)$. Each factor is nonnegative. Moreover, $1 - m_{-g} > 0$ because every $\sigma_h > 1$, and $\nu - 1 > 0$ exactly when $\sigma_0 < 1$. Thus $B \geq 0$, with equality only when $\nu = 1$, that is, at $\sigma_0 = 1$. □

Proposition 4 (Collective profitability). *For* $\sigma_0 < 1$, *industry* g*'s revenue and aggregate profit increase strictly, and every other industry's revenue and aggregate profit decrease strictly. At* $\sigma_0 = 1$, *they remain unchanged in every industry.*

Proof of Proposition 4. Let E_g and E_{-g} denote post-regulation expenditure on industry g and on all other industries combined. Two equilibrium conditions determine them.

First, wages exhaust revenue net of profit, and labor is supplied inelastically at wage w:

$$wL = (1 - m_g)E_g + (1 - m_{-g})E_{-g}.$$

Second, the outer nest fixes the expenditure ratio:

$$\frac{E_g}{E_{-g}} = \frac{S_g^R}{1 - S_g^R} = \frac{S_g \nu}{1 - S_g}.$$

Substituting $E_{-g} = E_g(1 - S_g)/(S_g\nu)$ into the first condition and solving gives

$$E_g = \frac{wL}{(1 - m_g) + \dfrac{1 - S_g}{S_g \nu}(1 - m_{-g})},$$

$$E_{-g} = \frac{wL}{(1 - m_{-g}) + \dfrac{S_g \nu}{1 - S_g}(1 - m_g)}.$$

Expenditure E_g is strictly increasing in ν, while E_{-g} is strictly decreasing. Margins are constant, so profits $m_g E_g$ and $m_{-g} E_{-g}$ move with revenue.

For $\sigma_0 < 1$, $\nu > 1$ and both changes are strict. Every other industry $h \neq g$ has share $S_h^R = S_h/D$, so its change in log revenue is

$$-\ln D + \Delta \ln Y = \Delta \ln E_{-g} < 0,$$

the same for all $h \neq g$. At $\sigma_0 = 1$, $\nu = D = 1$, so revenue and profit remain unchanged. □

Theorem 2 (Who wins in general equilibrium). *The change in log revenue $\Delta \ln r(\psi)$ is strictly decreasing in ψ. Firm ψ gains revenue and profit if and only if*

$$\ln\big(1 + \varepsilon(\psi)\big) < \Delta \ln P_g + \frac{B}{\sigma_g - 1}.$$

The threshold strictly exceeds $\ln(1 + \varepsilon_0)$, so the most productive firms always gain.

Proof of Theorem 2. Monotonicity and the cutoff. In (24), B and $\Delta \ln P_g$ are common to all firms, while $\ln(1 + \varepsilon(\psi))$ is strictly increasing in ψ. Thus $\Delta \ln r(\psi)$ is strictly decreasing. Setting (24) equal to zero gives the cutoff

$$\ln(1 + \varepsilon^*) = \Delta \ln P_g + \frac{B}{\sigma_g - 1}.$$

Firms with $\varepsilon(\psi) < \varepsilon^*$ gain revenue, and firms above the cutoff lose. At the constant markup, profit r/σ_g moves with revenue.

The threshold exceeds minimum log incidence. Incidence ε is strictly increasing in ψ, and more than one cost level carries positive revenue. Therefore, $(1 + \varepsilon(\psi))^{1-\sigma_g} \leq (1 + \varepsilon_0)^{1-\sigma_g}$ at

every cost level, with strict inequality on a set of positive share. Averaging with the weights in the power mean (20) gives

$$\begin{aligned}\left(\frac{P_g^R}{P_g}\right)^{1-\sigma_g} &= \int_{\psi_{\min}}^{\infty} s(\psi)\big(1+\varepsilon(\psi)\big)^{1-\sigma_g}\, dG(\psi) \\ &< (1+\varepsilon_0)^{1-\sigma_g}.\end{aligned}$$

Raising both sides to the negative power $1/(1-\sigma_g)$ reverses the inequality. Taking logs gives $\Delta \ln P_g > \ln(1+\varepsilon_0)$. Since $B \geq 0$ by Lemma 1, the threshold strictly exceeds $\ln(1+\varepsilon_0)$.

The most productive firms gain. Evaluating (24) at $\psi_{\min}$ gives

$$\Delta \ln r(\psi_{\min}) = (\sigma_g - 1)\big[\Delta \ln P_g - \ln(1+\varepsilon_0)\big] + B > 0.$$

□

Corollary 2 (Every firm gains under proportional incidence). *Suppose $\sigma_0 < 1$ and let incidence be proportional, $\varepsilon(\psi) = \varepsilon > 0$ for every ψ. Then every firm in industry g gains revenue and profit, with $\Delta \ln r(\psi) = B > 0$ for every ψ. At $\sigma_0 = 1$ the gain vanishes, and the regulation is neutral.*

Proof of Corollary 2. With a common incidence, (20) gives $(P_g^R/P_g)^{1-\sigma_g} = (1+\varepsilon)^{1-\sigma_g}$, so $\Delta \ln P_g = \ln(1+\varepsilon) > 0$. Substituting into (24), the partial-equilibrium term vanishes and $\Delta \ln r(\psi) = B$ for every firm; by Lemma 1, $B > 0$ for $\sigma_0 < 1$ and $B = 0$ at $\sigma_0 = 1$. Profit is r/σ_g and moves with revenue. □

Proposition 5 (Welfare loss). *With a common margin ($m_g = m_{-g}$), $\Delta \ln W = -\Delta \ln P_0 < 0$. With heterogeneous margins, welfare decreases to first order in $\Delta \ln P_g$:*

$$\Delta \ln W^{(1)} = -\, S_g\, \Delta \ln P_g \left[1 - (1-\sigma_0)(1-S_g)\frac{m_g - m_{-g}}{1-\kappa}\right] < 0.$$

Proof of Proposition 5. Common margin. Equation (21) gives $\Delta \ln P_0 = \ln D/(1-\sigma_0)$. Since $W = Y/P_0$,

$$\Delta \ln W = \Delta \ln Y - \frac{\ln D}{1-\sigma_0}.$$

Under a common margin, (23) gives $\kappa^R = \kappa$, so $\Delta \ln Y = 0$. Hence $\Delta \ln W = -\ln D/(1-\sigma_0)$. This expression is strictly negative for $\sigma_0 < 1$ because $D > 1$. At $\sigma_0 = 1$, it converges to $-S_g\, \Delta \ln P_g < 0$.

Heterogeneous margins. Expand to first order in $u \equiv \Delta \ln P_g$. The definitions of ν and D give

$$\nu = e^{(1-\sigma_0)u} \approx 1 + (1-\sigma_0)u,$$
$$D = 1 - S_g + S_g\nu \approx 1 + S_g(1-\sigma_0)u.$$

Thus $\Delta \ln P_0 = \ln D/(1-\sigma_0) \approx S_g u$. Equation (22) gives

$$S_g^R - S_g = \frac{S_g(\nu - D)}{D} \approx S_g(1-S_g)(1-\sigma_0)u.$$

By (23), $\kappa^R - \kappa = (S_g^R - S_g)(m_g - m_{-g})$. The income response is therefore

$$\begin{aligned}\Delta \ln Y = \ln \frac{1-\kappa}{1-\kappa^R} &\approx \frac{\kappa^R - \kappa}{1-\kappa} \\ &= S_g(1-S_g)(1-\sigma_0)u\frac{m_g - m_{-g}}{1-\kappa}.\end{aligned}$$

Collecting terms in $\Delta \ln W = \Delta \ln Y - \Delta \ln P_0$ yields

$$\Delta \ln W^{(1)} = -\, S_g u \left[1 - (1-\sigma_0)(1-S_g)\frac{m_g - m_{-g}}{1-\kappa}\right].$$

The sign of the bracket. The bracket is positive. If $m_g \le m_{-g}$, it exceeds one. Otherwise,

$$(1-\sigma_0)(m_g - m_{-g}) \le m_g - m_{-g} \le 1 - m_{-g},$$

and

$$(1-S_g)(1-m_{-g}) \le 1 - \kappa.$$

The difference in the latter inequality is strictly positive:

$$1 - \kappa - (1-S_g)(1-m_{-g}) = S_g(1-m_g) > 0.$$

The subtracted term is therefore strictly below one, so the bracket is positive. □

Lemma B.2 (Bounds under an elastic nest)**.** *For $\sigma_0 > 1$:*

(i) $D \le 1$, *so* $-\ln D \ge 0$;

(ii) $\left|S_g^R - S_g\right| \le -\ln D$;

(iii) $|\Delta \ln Y| \le \dfrac{\bar{\mu}}{\min_h \sigma_h}(-\ln D)$, *where* $\bar{\mu} \equiv \max_h \sigma_h/(\sigma_h - 1)$ *is the largest markup. In particular,* $|\Delta \ln Y| \le -\ln D$ *whenever* $\sigma_h \ge 2$ *for every industry* h, *and* $\Delta \ln Y = 0$

under a common margin.

Proof. *(i)* Since $\sigma_0 > 1$ and $\Delta \ln P_g > 0$, $\nu = e^{(1-\sigma_0)\Delta \ln P_g} \leq 1$. Hence $D = 1 - S_g + S_g\nu \leq 1$.

(ii) Equation (22) gives

$$S_g - S_g^R = \frac{S_g(1-S_g)(1-\nu)}{D} \leq S_g(1-\nu),$$

where the inequality uses $D \geq 1 - S_g$. Also,

$$-\ln D = -\ln\bigl(1 - S_g(1-\nu)\bigr) \geq S_g(1-\nu).$$

Combining the two inequalities gives (ii).

(iii) By (23),

$$\begin{aligned} |\Delta \ln Y| &= \left|\ln \frac{1-\kappa}{1-\kappa^R}\right| \\ &\leq \frac{|\kappa^R - \kappa|}{\min\{1-\kappa,\, 1-\kappa^R\}} \\ &\leq \bar{\mu}\,|\kappa^R - \kappa|. \end{aligned}$$

The last inequality follows because $1-\kappa$ is the expenditure-weighted mean of the $1-m_h$ and never falls below $1/\bar{\mu}$. Moreover,

$$|\kappa^R - \kappa| = |S_g^R - S_g|\,|m_g - m_{-g}| \leq \frac{|S_g^R - S_g|}{\min_h \sigma_h},$$

because both margins lie below $1/\min_h \sigma_h$. Combining this inequality with (ii) gives the bound.

If $\sigma_h \geq 2$ for every h, then $\bar{\mu} \leq 2 \leq \min_h \sigma_h$, so the factor is at most one. Under a common margin, $\kappa^R = \kappa$ and $\Delta \ln Y = 0$. □

Proposition 6 (Industry expenditure decreases). *For $\sigma_0 > 1$, expenditure on industry g decreases strictly: $B < 0$.*

Proof of Proposition 6. The power-mean argument in the proof of Theorem 2 gives $\Delta \ln P_g > 0$. Thus $\nu = e^{(1-\sigma_0)\Delta \ln P_g} < 1$ and $D = 1 - S_g + S_g\nu \in (\nu, 1)$. It follows that $S_g^R/S_g = \nu/D < 1$, so the outflow $d \equiv S_g - S_g^R$ is strictly positive. Equation (23) gives

$$\kappa^R - \kappa = (S_g^R - S_g)(m_g - m_{-g}) = d(m_{-g} - m_g).$$

If $m_{-g} \leq m_g$, then $\kappa^R \leq \kappa$ and $\Delta \ln Y \leq 0$. The decline in expenditure share therefore implies $B = \Delta \ln S_g + \Delta \ln Y < 0$.

If $m_{-g} > m_g$, write

$$B = \ln\Big(1 - \frac{d}{S_g}\Big) - \ln\Big(1 - \frac{d(m_{-g} - m_g)}{1 - \kappa}\Big).$$

This expression is negative if and only if $S_g(m_{-g} - m_g) < 1 - \kappa$. Substituting $1 - \kappa = 1 - S_g m_g - (1 - S_g)m_{-g}$ reduces the condition to $m_{-g} < 1$. This holds because every $\sigma_h > 1$. □

Theorem 3 (Winners under an elastic outer nest). *Let $\sigma_0 > 1$ and $\sigma_g > \sigma_0$. Suppose incidence is regressive, with $\varepsilon(\psi_{\min}) = 0$. Under a common margin across industries, the most productive firm gains revenue and profit. Because $\Delta \ln r(\psi)$ is strictly decreasing in ψ, the winning firms form a top segment of the productivity distribution. With heterogeneous margins, the same conclusions hold if $\sigma_h \geq 2$ for every industry h.*

Proof of Theorem 3. The index increases. Incidence is positive on a set of positive share. The power-mean argument in the proof of Theorem 2 therefore gives $\Delta \ln P_g > \ln(1+\varepsilon_0) = 0$.

Evaluating (24) at $\psi_{\min}$, where $\varepsilon_0 = 0$, and substituting $\Delta \ln S_g = (1 - \sigma_0)\Delta \ln P_g - \ln D$ from (22) into $B = \Delta \ln S_g + \Delta \ln Y$ gives the identity

$$\Delta \ln r(\psi_{\min}) = (\sigma_g - \sigma_0)\Delta \ln P_g - \ln D + \Delta \ln Y. \tag{30}$$

Common margin. Equation (23) gives $\kappa^R = \kappa$, so $\Delta \ln Y = 0$. Since $\Delta \ln P_g > 0$, $\nu = e^{(1-\sigma_0)\Delta \ln P_g} < 1$ and $D = 1 - S_g + S_g\nu < 1$, so $-\ln D > 0$ and the identity (30) yields

$$\begin{aligned}\Delta \ln r(\psi_{\min}) &= (\sigma_g - \sigma_0)\Delta \ln P_g - \ln D \\ &> (\sigma_g - \sigma_0)\Delta \ln P_g > 0.\end{aligned}$$

Heterogeneous margins. If $\sigma_h \geq 2$ for every h, Lemma B.2(iii) gives $|\Delta \ln Y| \leq -\ln D$. Hence $-\ln D + \Delta \ln Y \geq 0$ in (30), and the same bound holds.

Single crossing. Profit is r/σ_g and increases with revenue. In (24), the index and B are common across firms, while $\ln(1 + \varepsilon(\psi))$ is strictly increasing. Thus $\Delta \ln r(\psi)$ is strictly decreasing. Since its value at $\psi_{\min}$ is positive, the winners are the segment of the distribution below the single crossing. □

Proposition A.1 (Reduction to cost-function space). *Suppose cost minimization is isolated for every firm. Firm i's problem is equivalent to the following two-stage problem. First, compute $C_i(q_i)$ as a function of own output and the given factor prices alone. Second, choose a point on the demand curve to maximize $p_i q_i - C_i(q_i)$. Industry equilibrium is*

characterized entirely by the cost functions $\{C_i\}$ *and demand. Every primitive source of firm heterogeneity enters only through* $\{C_i\}$.

Proof of Proposition A.1. Under isolation, r is invariant to the choices of any individual firm. Given q_i alone, the cost-minimization program is therefore well defined:

$$\min_{x_i}\{r \cdot x_i : f_i(x_i) \geq q_i\}.$$

Its value $C_i(q_i)$ is independent of (q_{-i}, x_{-i}).

Firm i's full problem is

$$\max_{q_i, x_i}\{p_i(q_i)q_i - r \cdot x_i : f_i(x_i) \geq q_i\}.$$

For any candidate output, cost minimization gives the optimal input choice. Substituting its value function reduces the problem to

$$\max_{q_i}\{p_i(q_i)q_i - C_i(q_i)\}.$$

Here, $p_i(q_i)$ captures the product-market side: residual demand, or the aggregator under monopolistic competition. It depends on rivals only through their outputs.

By Shephard's lemma, C_i recovers all information about f_i and r that is relevant to production. Every equilibrium of the primitive game therefore corresponds to an equilibrium of the game in which firms are endowed with $\{C_i\}$, and conversely. □

Corollary C.1 (Kink deformations, stated in Appendix C). *Suppose there is* $\bar{\omega} \in (0, 1)$ *such that* $\psi^R_\omega = \psi_\omega$ *for every* $\omega \in (0, \bar{\omega})$*, while* $\psi^R_\omega \geq \psi_\omega$ *elsewhere, with strict inequality on a set of positive measure. Then every firm in* $(0, \bar{\omega})$ *strictly gains.*

Proof of Corollary C.1. The deformation weakly raises marginal costs everywhere and strictly raises them on a set of positive measure. Lemma B.1 gives $\rho > 1$. On $(0, \bar{\omega})$, the cost ratio equals one, so Theorem 1 applies at every point of the segment. □

Corollary C.2 (Vanishing burden, stated in Appendix C). *Suppose (i) the aggregator increases,* $\rho > 1$*, and (ii)* $\lim_{\omega \downarrow 0} \psi^R_\omega/\psi_\omega = 1$*. Then there exists* $\bar{\omega} > 0$ *such that every firm in* $(0, \bar{\omega})$ *strictly gains.*

Proof of Corollary C.2. Condition (i) gives a fixed $\rho > 1$. By condition (ii), $\psi^R_\omega/\psi_\omega \to 1$ as $\omega \downarrow 0$. There is therefore some $\bar{\omega} > 0$ such that $\psi^R_\omega/\psi_\omega < \rho$ throughout $(0, \bar{\omega})$. Theorem 1 gives strict gains for every firm in that segment. □

C Kinks, Vanishing Burdens, and Exit

This appendix presents two stronger variants of the top-sparing deformation. Each is a corollary of Theorem 1, with proofs in Appendix B. Corollary 1 applies to both variants.

Corollary C.1 (Kink deformations: an untouched top)**.** *Suppose there is* $\bar{\omega} \in (0,1)$ *such that* $\psi^R_\omega = \psi_\omega$ *for every* $\omega \in (0, \bar{\omega})$*. Elsewhere, suppose* $\psi^R_\omega \geq \psi_\omega$*, with strict inequality on a set of positive measure. Then* every *firm in* $(0, \bar{\omega})$ *strictly gains revenue, profit, and market share.*

The entire untouched segment grows. Under rent-sharing, a wage floor binds only for firms paying below it, leaving marginal costs unchanged above the productivity cutoff (Appendix K). Likewise, a compliance rule that codifies practices already used by the most productive firms leaves their costs unchanged. Whenever incidence has a kink, zero up to a cutoff and positive beyond it, every firm on the untouched side wins.

Incidence can instead be positive at every firm and vanish as productivity approaches the top of the distribution. To illustrate, take a cost distribution with a fixed, strictly positive minimum marginal cost and fatten its tail. The Pareto family is the canonical example. Every firm's cost is higher, but the increase vanishes as marginal cost approaches its minimum, where the most productive firms lie:

$$\frac{\psi^R_\omega}{\psi_\omega} \to 1 \qquad \text{as } \omega \downarrow 0.$$

The aggregator increases because costs are higher everywhere (Lemma B.1).

I call deformations with an increasing aggregator and vanishing incidence at the top *shape-changing*. In the parametric case, the deformation is implemented exactly by changing the shape parameter. Appendix D gives a closed-form CES–Pareto illustration. The following corollary states the result.

Corollary C.2 (Vanishing burden)**.** *Suppose (i) the aggregator increases,* $\rho > 1$*, and (ii)* $\lim_{\omega \downarrow 0} \psi^R_\omega / \psi_\omega = 1$*. Then there exists* $\bar{\omega} > 0$ *such that every firm in* $(0, \bar{\omega})$ *strictly gains revenue, profit, and market share.*

Remark C.1 (Scope of the conditions). The corollary requires no parametric family for G, no stochastic-dominance ranking of ψ^R relative to ψ, and no functional form for demand beyond the monotonicity already assumed. Costs need not increase at every firm. By Lemma B.1, any cost-increasing regulation satisfies condition (i), but that condition also admits deformations that *lower* some firms' costs.

The conditions are stated directly in terms of two observables of the deformation: the change in the aggregate price environment and incidence at the top of the productivity distribution. They can therefore be checked without solving for equilibrium. Proposition B.1 in Appendix B gives the exact necessary-and-sufficient characterization and identifies which conditions can and cannot be relaxed. Appendix K derives ordered incidence from production primitives: a *uniform* labor tax generates it through differences in factor intensities.

D CES–Pareto Illustration

This appendix provides a closed-form derivation of the parametric illustration in Section 2.2. Demand is CES with elasticity $\sigma > 1$, so the aggregator is

$$P(\boldsymbol{\mu\psi}) = \left[\int_0^1 \left(\tfrac{\sigma}{\sigma-1}\,\psi_\omega\right)^{1-\sigma} d\omega\right]^{\frac{1}{1-\sigma}}.$$

Marginal costs follow a Pareto distribution, $\psi_\omega = \psi_0(1-\omega)^{-1/\theta}$, with $\theta > 0$. A lower θ implies a fatter right tail, as in Section 4.5.

Evaluating the power integral in the aggregator gives

$$\begin{aligned}\int_0^1 \psi_\omega^{1-\sigma}\, d\omega &= \psi_0^{1-\sigma}\int_0^1 (1-\omega)^{\frac{\sigma-1}{\theta}}\, d\omega \\ &= \psi_0^{1-\sigma}\frac{\theta}{\theta+\sigma-1}.\end{aligned}$$

The aggregator therefore equals

$$P(\boldsymbol{\mu\psi}) = \mu\,\psi_0\left[\frac{\theta}{\theta+\sigma-1}\right]^{\frac{1}{1-\sigma}},$$

and firm ω's relative cost is

$$\frac{\psi_\omega}{P(\boldsymbol{\mu\psi})} = \frac{(1-\omega)^{-1/\theta}}{\mu\left[\dfrac{\theta}{\theta+\sigma-1}\right]^{\frac{1}{1-\sigma}}}. \tag{31}$$

The numerator varies across firms, and the denominator summarizes the distribution's shape at the industry level. The scale parameter ψ_0 cancels between numerator and denominator, as in Proposition 2.

A marginal *decrease* in θ is a shape-changing regulation. Every firm's cost increases and dispersion widens, while the proportional cost increase vanishes at the left tail. These

properties satisfy conditions (i)–(ii) of Corollary C.2 in Appendix C.

Differentiating the logarithm of (31) with respect to θ gives

$$\frac{\partial}{\partial\theta}\ln\frac{\psi_\omega}{P}=\frac{\ln(1-\omega)}{\theta^2}+\frac{1}{\theta(\theta+\sigma-1)}.$$

The first term comes from the firm's own cost. The second comes from the aggregator, using

$$\frac{1}{1-\sigma}\frac{\partial}{\partial\theta}\ln\left[\frac{\theta}{\theta+\sigma-1}\right]=-\frac{1}{\theta(\theta+\sigma-1)}.$$

The derivative of log relative cost is positive if and only if $\ln(1-\omega) > -\theta/(\theta+\sigma-1)$. Because the regulation *lowers* θ, a positive derivative means that the firm's relative cost decreases. Thus, firm ω benefits if and only if $\omega < \bar{\omega}$, where

$$\bar{\omega}=1-\exp\left(-\frac{\theta}{\theta+\sigma-1}\right)\in(0,1).$$

The cutoff $\bar{\omega}$ is *decreasing* in σ and *increasing* in θ. As $\sigma\to\infty$, $\bar{\omega}\to 0$; as $\theta\to\infty$, $\bar{\omega}\to 1-e^{-1}\approx 0.63$.

When varieties are highly substitutable, expenditure, and hence the aggregator, is dominated by firms at the top of the productivity distribution. These firms bear little of the regulatory burden, so the aggregate price response $\rho = P^R/P$ is weak and few firms benefit. Low substitutability makes the aggregator respond more to the heavily burdened right tail of costs, enlarging the winning set.

E Deformations and Pareto Parameters

Assume marginal costs are Pareto, $\Pr(\psi > y) = (\psi_{\min}/y)^\theta$, with scale $\psi_{\min}$ and shape θ. The elasticity of incidence is constant:

$$1+\varepsilon(\psi)=(1+\varepsilon_0)\left(\frac{\psi}{\psi_{\min}}\right)^\beta,\qquad \beta>-1.$$

Here, ε_0 is the incidence of the most productive firms. The restriction $\beta > -1$ ensures that the deformation is increasing in ψ and that deformed costs remain Pareto.

Substituting the incidence schedule gives

$$\psi^R=(1+\varepsilon_0)\,\psi_{\min}\left(\frac{\psi}{\psi_{\min}}\right)^{1+\beta}.$$

The deformed distribution therefore satisfies

$$\Pr\left(\psi^R > y\right) = \left(\frac{(1+\varepsilon_0)\,\psi_{\min}}{y}\right)^{\frac{\theta}{1+\beta}}.$$

It is Pareto, with scale and shape

$$\psi^R_{\min} = (1+\varepsilon_0)\,\psi_{\min}, \qquad \theta^R = \frac{\theta}{1+\beta}.$$

The map $(\varepsilon_0, \beta) \mapsto (\psi^R_{\min}, \theta^R)$ is one-to-one. Here, ε_0 is the ratio of the deformed to the original scale minus one, and β is the ratio of the original to the deformed shape minus one. Each statement below therefore holds in both directions.

Proposition E.1 (Scale)**.** *Incidence is proportional ($\beta = 0$) if and only if the deformation changes only the scale parameter.*

Proof of Proposition E.1. If $\beta = 0$, then $\theta^R = \theta$ and $\psi^R_{\min} = (1+\varepsilon_0)\,\psi_{\min}$, so only the scale parameter changes.

Conversely, if only the scale parameter changes, then $\theta^R = \theta$ implies $1+\beta = 1$ and hence $\beta = 0$. Incidence is then $1+\varepsilon(\psi) = 1+\varepsilon_0$ at every cost level and is therefore proportional. □

Proposition E.2 (Shape)**.** *Incidence is regressive ($\beta > 0$) or progressive ($\beta < 0$) if and only if the deformation changes the shape parameter. Regressive incidence fattens the tail, with $\theta/(1+\beta) < \theta$, while progressive incidence thins it. The scale parameter changes by the factor $1+\varepsilon_0$.*

Proof of Proposition E.2. The elasticity of the incidence schedule is

$$\frac{d\ln(1+\varepsilon(\psi))}{d\ln\psi} = \beta.$$

Incidence is therefore regressive exactly when $\beta > 0$ and progressive exactly when $\beta < 0$: incidence that rises with cost falls with productivity.

Since $\theta^R = \theta/(1+\beta)$, the shape parameter changes if and only if $\beta \neq 0$. In the regressive case, $1+\beta > 1$, so $\theta^R < \theta$ and the tail fattens. In the progressive case, $1+\beta \in (0,1)$, so $\theta^R > \theta$ and the tail thins. In either case, $\psi^R_{\min} = (1+\varepsilon_0)\,\psi_{\min}$, so the scale parameter changes by the factor $1+\varepsilon_0$. □

Proposition E.3 (Pure shape)**.** *Among cost-increasing deformations, the shape parameter changes alone if and only if incidence is regressive and vanishes for the most productive firms.*

Proof of Proposition E.3. Only the shape parameter changes if and only if $\psi^R_{\min} = \psi_{\min}$, or equivalently, $\varepsilon_0 = 0$. Suppose the deformation is cost-increasing and $\varepsilon_0 = 0$. Then

$$1 + \varepsilon(\psi) = \left(\frac{\psi}{\psi_{\min}}\right)^{\beta} \geq 1 \qquad \text{for every } \psi \geq \psi_{\min},$$

which requires $\beta \geq 0$. The case $\beta = 0$ with $\varepsilon_0 = 0$ is the identity deformation, excluded by Definition 1. Hence $\beta > 0$: incidence is regressive and vanishes for the most productive firms.

Conversely, regressive incidence that vanishes for the most productive firms has $\beta > 0$ and $\varepsilon_0 = 0$. The scale parameter is therefore unchanged, and only the shape parameter changes.

A cost-increasing *progressive* deformation attains its largest incidence at $\psi_{\min}$, where the factor $(\psi/\psi_{\min})^{\beta}$ is largest when $\beta < 0$. Cost increase therefore requires $\varepsilon_0 > 0$, so the scale parameter changes along with the shape parameter. □

F Estimating the Pareto Shape

This appendix details the estimation of the Pareto shape parameter θ used in Section 4.5.

Sample. For each treated product, the sample is the estimation firms of Section 3, and size is each firm's 2011 revenue in the category, the baseline size of the gradient regressions.

Moment. Under Pareto marginal costs with shape θ and CES revenue $r \propto \psi^{1-\sigma_g}$, the log revenue distance from the most productive firm, $\ln(r_{\max}/r_i) = (\sigma_g - 1)\ln(\psi_i/\psi_{\min})$, is exponential with rate $\lambda = \theta/(\sigma_g - 1)$.

Estimator. I use the estimator of Hill (1975) with the most productive firm as the threshold and the other $n - 1$ sampled firms in the tail:

$$\hat{\lambda} \;=\; \Big[\frac{1}{n-1}\sum_{i \neq \max} \ln\frac{r_{\max}}{r_i}\Big]^{-1}, \qquad \hat{\theta} \;=\; (\sigma_g - 1)\,\hat{\lambda},$$

with $\sigma_g = 3.9$ as in Section 4.5. The estimator is the maximum-likelihood rate of the exponential moment, and its standard error is $\hat{\theta}/\sqrt{n-1}$. Table 7 reports the inputs and estimates by product.

Product	n	$r_{\max}$ ($M)	Mean $\ln(r_{\max}/r_i)$	$\hat{\theta}$ (SE)
Nuts, oil-roasted	24	152.6	3.27	0.89 (0.18)
Nuts, dry-roasted	34	152.6	3.39	0.86 (0.15)
Dried fruit, true-dried	20	145.3	3.64	0.80 (0.18)
Peanut butter, creamy	14	254.8	3.90	0.74 (0.21)
Peanut butter, chunky	12	254.8	3.67	0.79 (0.24)
Chocolate, nut/PB	35	1283.5	5.13	0.57 (0.10)
Granola bars, w/ inclusions	15	279.8	3.91	0.74 (0.20)
Granola cereal	12	32.0	2.31	1.26 (0.38)
Cereal	25	1189.2	5.35	0.54 (0.11)
Seasoning	35	257.6	4.51	0.64 (0.11)
Crackers, flavored	23	463.8	5.12	0.57 (0.12)

Table 7: Pareto shape estimates by treated product. n is the number of estimation firms, $r_{\max}$ is the 2011 revenue of the largest firm in the sampled stores, and the mean log distance is taken over the other $n-1$ firms. $\hat{\theta} = (\sigma_g - 1)\hat{\lambda}$ with $\sigma_g = 3.9$; standard errors are $\hat{\theta}/\sqrt{n-1}$.

G Sensitivity of the Quantification

This appendix reports standard errors for the objects calibrated in Section 4.5 and the results under alternative elasticities.

Standard errors. The price-index change (28) is a closed form in β, with $\partial \Delta \ln P_g / \partial \beta = 1/[\theta + (1+\beta)(\sigma_g - 1)]$, so the delta method gives its standard error from that of β, holding the calibrated parameters fixed. The welfare cost inherits the standard error through (25), and the total adds variances across products. Table 8 reports the results.

Elasticities. Table 9 reports the welfare cost and the share of firms gaining under $\sigma_0 \in \{0.5, 1, 2\}$ and under the interquartile range of the within-industry elasticity in Hottman et al. (2016), $\sigma_g \in \{3.1, 5.1\}$. The cost is insensitive to σ_0: at these expenditure shares it equals the share times the price-index rise at any outer elasticity. The share of firms gaining decreases as σ_0 increases, because expenditure leaves the industry (Section 4.6). The within-industry elasticity moves the cost directly through (28).

Incidence for the most productive firms. A proportional component $1+\varepsilon_0$ adds $\ln(1+\varepsilon_0)$ to $\Delta \ln P_g$ and therefore approximately $S_g \ln(1+\varepsilon_0)$ to each product's cost, leaving the partial-equilibrium cutoff unchanged.

Product	β (SE)	$\hat{\theta}$ (SE)	$\Delta \ln P_g$ (%) (SE)	Welfare cost ($M/yr)	95% CI
Nuts, oil-roasted	0.456 (0.092)	0.89 (0.18)	10.3 (1.8)	41	[27, 56]
Nuts, dry-roasted	0.238 (0.077)	0.86 (0.15)	5.8 (1.7)	44	[18, 70]
Dried fruit, true-dried	0.292 (0.096)	0.80 (0.18)	7.1 (2.1)	19	[8, 31]
Peanut butter, creamy	0.103 (0.039)	0.74 (0.21)	2.7 (1.0)	26	[7, 45]
Peanut butter, chunky	0.097 (0.050)	0.79 (0.24)	2.5 (1.3)	5	[0, 11]
Chocolate, nut/PB	0.039 (0.028)	0.57 (0.10)	1.1 (0.8)	27	[-10, 64]
Granola bars, w/ inclusions	0.348 (0.188)	0.74 (0.20)	8.4 (4.0)	56	[2, 111]
Granola cereal	0.256 (0.120)	1.26 (0.38)	5.7 (2.5)	12	[2, 22]
Cereal	0.064 (0.019)	0.54 (0.11)	1.8 (0.5)	105	[45, 164]
Seasoning	0.049 (0.018)	0.64 (0.11)	1.4 (0.5)	15	[4, 26]
Crackers, flavored	0.027 (0.048)	0.57 (0.12)	0.8 (1.4)	11	[-26, 47]
Total				362	[258, 466]

Table 8: Uncertainty in the quantification. β and its standard error are from Section 3; $\hat{\theta}$ and its standard error are from Appendix F and are reported as inputs. The standard error of $\Delta \ln P_g$ and the confidence interval of the welfare cost propagate the standard error of β by the delta method, with θ, $\sigma_g = 3.9$, $\sigma_0 = 0.5$, and S_g held fixed. The total's interval adds variances across products.

	$\sigma_g = 3.9$			$\sigma_0 = 0.5$	
Product	$\sigma_0 = 0.5$	$\sigma_0 = 1$	$\sigma_0 = 2$	$\sigma_g = 3.1$	$\sigma_g = 5.1$
Nuts, oil-roasted	41 (21)	40 (18)	38 (12)	58 (22)	29 (20)
Nuts, dry-roasted	44 (22)	44 (19)	42 (13)	61 (23)	31 (21)
Dried fruit, true-dried	19 (20)	19 (18)	18 (12)	27 (21)	13 (20)
Peanut butter, creamy	26 (21)	26 (18)	25 (12)	36 (22)	18 (20)
Peanut butter, chunky	5 (21)	5 (19)	5 (13)	8 (23)	4 (21)
Chocolate, nut/PB	27 (17)	27 (15)	27 (10)	37 (18)	19 (16)
Granola bars, w/ inclusions	56 (19)	55 (16)	53 (11)	79 (20)	40 (18)
Granola cereal	12 (28)	12 (24)	12 (17)	17 (29)	9 (27)
Cereal	105 (16)	104 (14)	103 (10)	145 (17)	74 (16)
Seasoning	15 (19)	15 (16)	15 (11)	21 (20)	11 (18)
Crackers, flavored	11 (17)	11 (15)	11 (10)	15 (18)	8 (17)
Total	362	358	350	502	255

Table 9: Welfare cost ($M/yr) and, in parentheses, the share of firms gaining (%), under alternative elasticities. The first column repeats the baseline of Table 5.

H The Mechanism Beyond HSA

The results in Section 2 rely on two properties of HSA demand. The first is *equivalence*: maximizing over price is equivalent to maximizing over relative price, and the optimum depends on the firm's cost only through ψ_ω/P. The second is *scaling*: equilibrium profit equals expenditure Y times a decreasing function of that ratio. Because neither the firm nor the regulation changes Y, a fall in relative cost raises absolute profit.

This appendix examines how far Propositions 1 and 2 extend when these properties weaken. The analysis begins with the other two homothetic classes of Matsuyama and Ushchev (2017), then considers other demand systems. Notation is local to this appendix: $\mathcal{I}(\mathbf{p})$ denotes a class's ideal price index, or unit expenditure function, and $\mathcal{B}(\mathbf{p})$ and $\mathcal{C}(\mathbf{p})$ denote its secondary aggregators.

H.1 The other homothetic classes

Matsuyama and Ushchev (2017) characterize three pairwise disjoint classes of homothetic demand systems whose only common member is CES. HSA is one of them.

The second class, *homothetic with direct implicit additivity* (HDIA), defines utility implicitly by $\int \phi_\omega(x_\omega/U)\,d\omega = 1$, with each ϕ_ω increasing and concave. Demand is

$$x_\omega = \frac{Y}{\mathcal{I}(\mathbf{p})}(\phi'_\omega)^{-1}\left(\frac{p_\omega}{\mathcal{B}(\mathbf{p})}\right).$$

The aggregator $\mathcal{B}$ is determined by the adding-up condition

$$\int \phi_\omega\Big((\phi'_\omega)^{-1}\Big(\frac{p_\omega}{\mathcal{B}}\Big)\Big)\,d\omega = 1.$$

The third class, *homothetic with indirect implicit additivity* (HIIA), defines the ideal price index through the adding-up condition

$$\int \theta_\omega\Big(\frac{p_\omega}{\mathcal{I}}\Big)\,d\omega = 1.$$

Demand is

$$x_\omega = \frac{Y\,\theta'_\omega(p_\omega/\mathcal{I})}{\mathcal{C}(\mathbf{p})}, \qquad \mathcal{C} = \int p_\nu\,\theta'_\nu(p_\nu/\mathcal{I})\,d\nu.$$

In both classes, an atomistic firm's profit factors into a scaling term common to all firms

and a function of one relative-cost statistic:

$$\begin{aligned}\pi_\omega &= \frac{Y\mathcal{B}}{\mathcal{I}}\left(z_\omega - \frac{\psi_\omega}{\mathcal{B}}\right)(\phi'_\omega)^{-1}(z_\omega) \quad \text{(HDIA)},\\ \pi_\omega &= \frac{Y\mathcal{I}}{\mathcal{C}}\left(z_\omega - \frac{\psi_\omega}{\mathcal{I}}\right)\theta'_\omega(z_\omega) \qquad \text{(HIIA)}.\end{aligned} \tag{32}$$

Here, $z_\omega = p_\omega/\mathcal{B}$ under HDIA and $z_\omega = p_\omega/\mathcal{I}$ under HIIA.

The equivalence therefore extends to both classes. The firm's cost enters its problem only through $\psi_\omega/\mathcal{B}$ under HDIA or $\psi_\omega/\mathcal{I}$ under HIIA. Under HIIA, the sufficient statistic is cost relative to the ideal price index itself. The scaling property does not extend automatically: the aggregators in $\mathcal{B}/\mathcal{I}$ and $\mathcal{I}/\mathcal{C}$ move independently. By Matsuyama and Ushchev (2017, Propositions 2 and 3), these ratios are constant only under CES.

Lemma H.1 (Aggregator properties)**.** *The aggregators $\mathcal{B}$ under HDIA and $\mathcal{I}$ under HIIA exist, are unique, and are linearly homogeneous. Each increases when prices increase on a set of positive measure and decrease nowhere.*

Proof. Each aggregator is defined by an adding-up condition whose integrand moves monotonically in opposite directions with the price and the aggregator. Under HDIA, $(\phi')^{-1}$ is decreasing, so the integrand decreases with price and increases with $\mathcal{B}$. Under HIIA, θ is increasing, so the integrand increases with price and decreases with $\mathcal{I}$.

The integral crosses its target exactly once, giving existence and uniqueness. Scaling all prices and the aggregator by the same factor leaves every ratio unchanged, giving linear homogeneity. A price increase on a set of positive measure, with no price decreases, moves the integral away from its target. Only an increase in the aggregator restores the adding-up condition, giving monotonicity. □

Proposition H.1 (Regressive and proportional incidence beyond HSA)**.** *The following statements hold under both HDIA and HIIA.*

(i) Under regressive incidence as in Proposition 1, the sufficient statistic strictly decreases for the most productive firms and strictly increases for the least productive firms. Every relative outcome—relative price, relative quantity, and profit relative to any rival—therefore improves for the former and deteriorates for the latter.

(ii) If, in addition, the deformation raises the competition aggregator proportionally at least as much as the ideal price index,

$$\begin{aligned}\Delta \ln \mathcal{B} \geq \Delta \ln \mathcal{I} \quad &\textit{under HDIA},\\ \Delta \ln \mathcal{I} \geq \Delta \ln \mathcal{C} \quad &\textit{under HIIA},\end{aligned}$$

then the most productive firms gain in absolute profit.

(iii) A proportional deformation, $\psi_\omega^R = (1+\varepsilon)\psi_\omega$, leaves every firm's revenue, share, and profit unchanged.

Proof. (i) Lemma H.1 supplies exactly the properties used in the proof of Proposition 1. That proof therefore applies with the class's aggregator in place of P. Every firm's cost increases by at least the factor faced by the most productive firms, and some costs increase by strictly more. The aggregator therefore increases by strictly more than that factor, lowering the sufficient statistic at the top. The corresponding bound gives the result for the least productive firms.

(ii) By (32) and the envelope theorem, equilibrium profit equals the scaling term, $Y\mathcal{B}/\mathcal{I}$ or $Y\mathcal{I}/\mathcal{C}$, times a strictly decreasing function of the sufficient statistic. Under the stated condition, the scaling term weakly increases. By part (i), the value of the decreasing function strictly increases for the most productive firms. Their absolute profit therefore increases.

(iii) Consider a candidate equilibrium in which every price scales by $1+\varepsilon$. Homogeneity leaves every sufficient statistic, $\psi_\omega/\mathcal{B}$ or $\psi_\omega/\mathcal{I}$, unchanged. Every optimal relative price is therefore unchanged, so the candidate is a fixed point. Both aggregators scale by $1+\varepsilon$, leaving the scaling ratios unchanged. Equation (32) then gives unchanged profit for every firm. □

Remark H.1 (When the level condition holds). The condition in part (ii) holds with equality for proportional deformations. It holds identically in the symmetric Kimball (1995) case, where $\mathcal{B}$ and $\mathcal{I}$ are proportional in symmetric equilibrium, and under CES.

To first order, both sides of the condition are weighted averages of equilibrium price responses. The ideal price index uses expenditure shares as weights; the competition aggregator uses the weights from its own adding-up condition. The condition therefore requires the aggregator governing competition to place at least as much weight on the burdened firms as the cost-of-living index does.

H.2 Other demand systems

Translog demand. The symmetric translog of Feenstra (2003) belongs to HSA, with the geometric mean of prices as its aggregator. The results of Section 2 therefore apply without modification. The general asymmetric translog lies outside HSA: budget shares depend on every price through pairwise coefficients that cannot be collapsed into a single ratio. The equivalence therefore fails at the first step.

Almost ideal demand. The AIDS system of Deaton and Muellbauer (1980) is non-homothetic by design. Real income enters budget shares, so no aggregator of prices alone summarizes a firm's position. The framework therefore does not apply.

Additively separable preferences. In the variable-elasticity class of Zhelobodko et al. (2012), demand is

$$x_\omega = (u')^{-1}(\lambda p_\omega),$$

where λ is the marginal utility of income. Taking λ as given, the firm's problem depends on its cost through $\lambda\psi_\omega$. A regressive deformation moves this statistic in the direction described in the paper: λ increases with the price profile while the most productive firms' costs barely move.

The marginal utility of income λ depends on income as well as prices. Unless preferences are homothetic, it is not homogeneous of degree minus one in prices. A proportional deformation therefore changes $\lambda\psi_\omega$ across firms: Proposition 2 fails, and proportional cost increases redistribute.

Linear demand. In Melitz and Ottaviano (2008), maximized profit is $(\bar{p} - \psi_\omega)^2/(8b)$, where $\bar{p}$ is the demand intercept. The sufficient statistic is the cost *difference* $\bar{p} - \psi_\omega$, and the taxonomy changes accordingly. A proportional deformation is not neutral because the intercept is a preference parameter that does not scale with costs.

Under Proposition 3, an additive deformation redistributes against the most productive firms. Here, it subtracts a common constant from every firm's $\bar{p} - \psi_\omega$, a proportionally smaller loss for the most productive firms. The additive case therefore reverses sign relative to the homothetic taxonomy.

Nested CES. When elasticities differ across nests, nested CES belongs to none of the three classes. A firm's position depends on two statistics: its cost relative to the nest index and the nest index relative to the economy's price index. Section 4 analyzes this system directly.

H.3 Summary

I The Estimating Equation with Endogenous Expenditure

This appendix extends the derivation in Section 2.6 to allow expenditure and the price of the input bundle to vary across periods. Let Y^p and e^p denote their values in period $p \in \{\text{pre}, \text{post}\}$, and let P^p denote that period's price index. Collect the terms in (8) that

Demand system	Equivalence	Absolute gains	Neutrality	Sufficient statistic
CES	Yes	Yes	Yes	ψ/P
Symmetric translog	Yes	Yes	Yes	ψ/P
Kimball	Yes	Yes (symmetric)	Yes	$\psi/\mathcal{B}$
HDIA	Yes	Conditional	Yes	$\psi/\mathcal{B}$
HIIA	Yes	Conditional	Yes	$\psi/\mathcal{I}$
Nested CES	Within nest	Section 4	Section 4	ψ/P_g and P_g/P_0
Additively separable	Given λ	Directional	No	$\lambda\psi$
Asymmetric translog	No	—	—	None
Linear demand	No	Additive sign reversal	No	$\bar{p}-\psi$
AIDS	No	—	—	None

Table 10: The mechanism across demand systems. Equivalence means that the firm's problem depends on its cost through a single statistic. Absolute gains mean that regressive incidence raises the most productive firms' absolute profit, as in Proposition 1. For HDIA and HIIA, these gains hold under the condition in Proposition H.1(ii). Neutrality means that proportional cost increases leave revenue, market share, and profit unchanged, as in Proposition 2.

are common to all firms into

$$A^p = \ln Y^p + (\sigma-1)\ln\frac{P^p}{\mu\, e^p}.$$

Equations (8) and (16) give pre-regulation log revenue:

$$\ln r_\omega^{\text{pre}} = \left(A^{\text{pre}} - A^0\right) + \ln r_\omega^0.$$

Under the regulation, incidence has constant elasticity:

$$(\sigma-1)\ln(1+\varepsilon_\omega) = (\sigma-1)c - \tau\left(\ln r_\omega^0 - A^0\right).$$

Substituting gives post-regulation log revenue:

$$\ln r_\omega^{\text{post}} = \left(A^{\text{post}} - A^0 - (\sigma-1)c - \tau A^0\right) + (1+\tau)\ln r_\omega^0.$$

All terms other than $\ln r_\omega^0$ are common to firms within a period. Combining the two periods using the indicator post_t and grouping terms as in Section 2.6 yields

$$\ln r_{\omega t} = \beta\left(\ln r_\omega^0 \times \text{post}_t\right) + \delta_\omega + \delta_t, \qquad \beta = \tau.$$

The firm and time effects are

$$\begin{aligned}\delta_\omega &= \ln r^0_\omega,\\ \delta_t &= \left(A^{\text{pre}} - A^0\right)\\ &\quad + \left(A^{\text{post}} - A^{\text{pre}} - (\sigma-1)c - \tau A^0\right)\text{post}_t.\end{aligned}$$

This is the estimating equation (18). Movements in expenditure, the input price, and the price index affect only the period constants, which the time effects absorb. The interaction coefficient remains the elasticity of incidence.

J The Gradient in Two Benchmark Models

Section 2.5 states that proportional incidence and fixed-cost selection leave the productivity–size gradient unchanged. This appendix derives that invariance in two benchmark models. In a competitive industry producing a homogeneous good, the gradient is fixed by technology; in the canonical selection model of differentiated varieties, fixed costs act only through the exit cutoff.

J.1 A competitive industry: Hopenhayn (1992)

A single competitive industry produces a homogeneous good. The output price p and the wage w are determined in equilibrium, and individual firms take both as given. A firm's state is its productivity $\varphi > 0$, which evolves according to a Markov transition $\nu(\cdot \mid \varphi)$, independently across firms. Each period, a firm with productivity φ chooses labor $\ell \geq 0$ to maximize static profit:

$$\pi(\varphi, p, w) = \max_{\ell \geq 0}\left\{p\, f(\varphi, \ell) - w\,\ell\right\} - c_f.$$

The production function f is strictly increasing and concave in ℓ, and $c_f > 0$ is a fixed operating cost paid each period. The first-order condition $p\, f_\ell(\varphi, \ell^*) = w$ determines optimal labor.

The output price satisfies $p = D(Q)$, where Q is total industry output and D is strictly decreasing. The wage satisfies $w = W(N)$, where N is total labor demand and W is weakly increasing. These two market-clearing functions are the only objects through which the rest of the economy enters the industry.

At the start of each period, a firm decides whether to continue or exit at zero scrap value.

Its continuation value is

$$V(\varphi) = \pi(\varphi, p, w) + \beta_d \int \max\{V(\varphi'), 0\}\, \nu(d\varphi' \mid \varphi),$$

where β_d is the discount factor. Under standard monotonicity assumptions on ν, this value defines an exit threshold. Potential entrants pay a sunk cost and draw initial productivity from a fixed distribution. Entry continues while the expected value covers the cost.

Specialize the technology to $f(\varphi, \ell) = \varphi\, \ell^{\alpha}$, with $\alpha \in (0, 1)$. Productivity scales output multiplicatively, and firms share the same labor elasticity. The first-order condition gives $\ell^* = (\alpha p \varphi / w)^{1/(1-\alpha)}$. Substituting into revenue $r = p\, \varphi\, (\ell^*)^{\alpha}$ yields

$$\begin{aligned} \ln r(\varphi, p, w) &= K + \frac{1}{1-\alpha} \ln p + \frac{1}{1-\alpha} \ln \varphi - \frac{\alpha}{1-\alpha} \ln w, \\ K &= \frac{\alpha}{1-\alpha} \ln \alpha. \end{aligned} \tag{33}$$

This equation has the same structure as (8). Log revenue is linear in log productivity, with a gradient $1/(1-\alpha)$ fixed by technology. The price and wage enter the intercept, which is common to all firms.

The first-order condition also gives $w\, \ell^* = \alpha r$. Every firm therefore has the same labor cost share, regardless of productivity, and profit is $\pi = (1-\alpha) r - c_f$.

Proposition J.1 (The gradient cannot move)**.** *Under the technology above, any shift $(p, w) \to (p', w')$ produces the same change in log revenue for every firm:*

$$\Delta \ln r = \frac{\Delta \ln p}{1-\alpha} - \frac{\alpha\, \Delta \ln w}{1-\alpha}.$$

The gradient of log revenue with respect to log productivity equals $1/(1-\alpha)$ before and after the shift. Changes in demand D, labor supply W, or the fixed cost c_f, individually or jointly, leave the gradient unchanged.

Proof. Differencing (33) at fixed φ eliminates the productivity term, so $\Delta \ln r$ is independent of φ. The slope of (33) in $\ln \varphi$ is $1/(1-\alpha)$ at any (p, w). The fixed cost enters neither revenue nor its difference. □

Corollary J.1 (Survivors gain or lose together)**.** *Revenue increases for all surviving firms if and only if $p'/p > (w'/w)^{\alpha}$, and decreases for all of them otherwise. Profit changes have the same sign:*

$$\Delta \pi(\varphi) = (1-\alpha)\, \Delta r(\varphi),$$

because the fixed cost drops out of the difference. No combination of shifts in D and W makes some surviving firms gain while others lose.

Proof. Revenue increases if and only if $\Delta \ln r > 0$, or equivalently, $\ln(p'/p) > \alpha \ln(w'/w)$. This condition is independent of φ, so it holds for all firms or none. Since $\pi = (1-\alpha)r - c_f$ and c_f is unchanged, profit changes have the same sign as revenue changes. □

The magnitude of the change varies across firms. The exact revenue difference is

$$\Delta r(\varphi) = \alpha^{\alpha/(1-\alpha)}\, \varphi^{1/(1-\alpha)}\, \gamma,$$

where γ collects the price and wage terms and is independent of φ. Thus, $|\Delta r(\varphi)| \propto \varphi^{1/(1-\alpha)}$: more productive firms experience larger absolute changes in the same direction as everyone else.

A regulation can shift demand or labor supply, impose a wage floor, or raise the fixed cost. A wage floor above the market-clearing level raises the exit threshold, lowers output, and raises the output price. Any combination of these changes reaches firms only through (p, w) and the exit margin. Survivors' log revenues move in parallel, the exit threshold moves, and the gradient remains unchanged. In the estimating equation of Section 2.6, every response appears in the time effects and the survivor set. Here, $\beta = 0$ is a theorem.

J.2 The selection model: Melitz (2003)

Demand follows the CES system of Section 2.4. A representative consumer has elasticity $\sigma > 1$ over a continuum of varieties. With aggregate expenditure Y and price index P, a variety priced at $p(\omega)$ earns revenue $r(\omega) = Y\,(p(\omega)/P)^{1-\sigma}$.

Each firm draws productivity φ from a distribution G on $(0, \infty)$. Production requires a fixed cost $F > 0$ in units of labor and q/φ units of labor for q units of output. With the wage as numeraire, marginal cost is $1/\varphi$. Every firm charges the CES markup, $p(\varphi) = \frac{\sigma}{\sigma-1}\frac{1}{\varphi}$, so revenue is a power function of productivity:

$$\begin{aligned} r(\varphi) &= r_1\, \varphi^{\sigma-1}, \\ r_1 &\equiv Y \left(\frac{\sigma}{\sigma-1}\right)^{1-\sigma} P^{\sigma-1}. \end{aligned} \tag{34}$$

The factor r_1 is common to all firms.

Operating profit is $\pi(\varphi) = r(\varphi)/\sigma - F$. A firm produces if and only if operating profit is

non-negative, which defines the cutoff φ^* by

$$\frac{r_1(\varphi^*)^{\sigma-1}}{\sigma} = F.$$

Entrants pay a sunk cost F_e to draw from G. Free entry sets expected profit net of the entry cost to zero:

$$[1 - G(\varphi^*)]\,\bar{\pi}(\varphi^*) = F_e.$$

The equilibrium consists of the cutoff φ^*, the mass of active firms M, and the price index P. These are determined by the zero-cutoff-profit condition, free entry, the price index over survivors, and labor-market clearing.

Under a Pareto distribution, $G(\varphi) = 1 - (\varphi_{\min}/\varphi)^k$ with $k > \sigma - 1$, these objects have closed forms:

$$\begin{aligned}
\bar{\pi} &= \frac{F(\sigma-1)}{k-(\sigma-1)}, \\
\varphi^* &= \varphi_{\min}\left(\frac{k-(\sigma-1)}{\sigma-1}\frac{F_e}{F}\right)^{-1/k}, \\
M &= \frac{L}{F\,\sigma k/(k-(\sigma-1))}.
\end{aligned}$$

Proposition J.2 (Fixed costs move the cutoff, not the gradient)**.** *Raising the fixed cost F raises the cutoff φ^* and induces the least productive survivors to exit. For the firms that remain,*

$$\ln r = \ln r_1 + (\sigma - 1)\ln\varphi.$$

The common factor r_1 adjusts through P and M, while the gradient is $\sigma - 1$ before and after. Relative revenues are invariant:

$$\frac{r(\varphi_1)}{r(\varphi_2)} = \left(\frac{\varphi_1}{\varphi_2}\right)^{\sigma-1} \qquad \textit{for any two survivors,}$$

whatever happens to P, M, Y, or f. No surviving firm's relative position changes. A fixed-cost regulation creates winners only through the extensive margin.

Proof. Under the Pareto parameterization, the cutoff

$$\varphi^* = \varphi_{\min}\left(\frac{k-(\sigma-1)}{\sigma-1}\frac{F_e}{F}\right)^{-1/k}$$

is strictly increasing in F.

The revenue-ratio result requires no distributional assumption. By (34), $r(\varphi_1)/r(\varphi_2) = (\varphi_1/\varphi_2)^{\sigma-1}$, which is independent of r_1. The slope of $\ln r$ in $\ln\varphi$ is $\sigma - 1$ at any value of the

aggregates. □

The exit margin is the amplification channel of Corollary 1. Exiting firms shrink the set over which the price index aggregates. The index increases, and survivors absorb the exiting firms' demand in proportion to their shares. Relative productivities and the common markup pin down relative prices. Reallocation among survivors therefore operates only through the common index, with no change in relative prices.

A regulation that moves the gradient must therefore change marginal costs asymmetrically across firms, as in the deformation of Section 2.2. As in the competitive benchmark, a fixed-cost regulation enters the time effects and the survivor set in the estimating equation. Again, $\beta = 0$.

K Microfoundations of Regressive Incidence

The theorems in Section 2 take the deformation of marginal costs as a primitive. This appendix derives regressive incidence and a kink from production primitives. A uniform labor tax generates regressive incidence through differences in factor intensity; a wage floor generates a kink under rent-sharing. Both constructions apply to any regulated input priced at a common factor price.

K.1 Heterogeneous factor intensity: regressive incidence

Let firm φ produce with the constant-returns technology $q = \varphi K^{\alpha(\varphi)} L^{1-\alpha(\varphi)}$. The capital share $\alpha(\varphi)$ is *increasing* in productivity, so larger, more productive firms are more capital-intensive. Cost minimization gives the Cobb–Douglas unit cost

$$\psi(\varphi) = \frac{\chi\big(\alpha(\varphi)\big)\, r_K^{\alpha(\varphi)}\, w^{1-\alpha(\varphi)}}{\varphi},$$
$$\chi(\alpha) = \alpha^{-\alpha}(1-\alpha)^{-(1-\alpha)}.$$

The elasticity of unit cost with respect to the wage equals the firm's labor cost share:

$$\frac{\partial \ln \psi(\varphi)}{\partial \ln w} = 1 - \alpha(\varphi).$$

A *uniform* labor tax t raises the price of labor from w to $(1+t)w$. Its incidence on unit cost is therefore

$$1 + \varepsilon(\varphi) = (1+t)^{1-\alpha(\varphi)}.$$

Because $\alpha(\cdot)$ is increasing, incidence is strictly decreasing in φ. A uniform regulation therefore generates regressive incidence. For small t, $\varepsilon(\varphi) \approx \big(1-\alpha(\varphi)\big)t$.

When $\alpha(\varphi) \uparrow \bar{\alpha} \leq 1$, the burden at the top of the productivity distribution is

$$\varepsilon_0 = (1+t)^{1-\bar{\alpha}} - 1 \geq 0.$$

Proposition 1 applies whenever $\alpha(\cdot)$ is not constant. Each firm operates under constant returns and is isolated, as in Proposition A.1. The construction therefore belongs to the class of models in Appendix A.

K.2 Rent-sharing wages: a kink

Alternatively, let all firms use the technology $q = \varphi L$ and pay productivity-dependent wages

$$w(\varphi) = w_0\, \varphi^{\gamma}, \qquad \gamma \in (0,1).$$

This is the rent-sharing schedule derived from bargaining in Helpman et al. (2010), under which more productive firms pay higher wages. Marginal cost is

$$\psi(\varphi) = \frac{w(\varphi)}{\varphi} = w_0\, \varphi^{\gamma-1}.$$

A wage floor $\bar{w}$ binds if and only if $w(\varphi) < \bar{w}$, which occurs below the productivity cutoff $\bar{\varphi} = (\bar{w}/w_0)^{1/\gamma}$. Post-regulation marginal cost is

$$\psi^R(\varphi) = \begin{cases} \bar{w}/\varphi & \varphi < \bar{\varphi} \quad \text{(floor binds)}, \\ \psi(\varphi) & \varphi \geq \bar{\varphi} \quad \text{(unchanged)}. \end{cases}$$

This is a continuous deformation with two pieces: the burden is zero above the cutoff and strictly positive below it. It is the kink of Corollary C.1, so every firm above $\bar{\varphi}$ strictly gains. Because each atomistic firm takes its wage schedule as given, the construction is isolated (Definition 2) and remains in cost-function space.

L Cross-Industry Substitution: Nested CES in Partial Equilibrium

This appendix extends the partial-equilibrium analysis of Section 2 to allow substitution across industries. Consumers allocate expenditure across industries g with an outer elas-

ticity σ_0 and across firms within each industry with inner elasticities $\sigma_g > 1$. Aggregate expenditure Y is held fixed, while the industry's expenditure share S_g captures substitution across industries.

Firm j in industry g earns profit

$$\pi_j^g = \frac{Y S_g}{\sigma_g} \left(\frac{p_j}{P_g} \right)^{1-\sigma_g},$$

where the industry's expenditure share is

$$S_g = \frac{\alpha_g P_g^{1-\sigma_0}}{\sum_h \alpha_h P_h^{1-\sigma_0}}.$$

When the regulation raises the industry price index P_g, expenditure shifts toward other industries.

Proposition L.1 (Gains with cross-industry substitution)**.** *Consider a regulation with regressive incidence and vanishing burden in industry g. The most productive firms gain in absolute profit if and only if*

$$\sigma_g - 1 > (\sigma_0 - 1)(1 - S_g).$$

A sufficient condition is $\sigma_g > \sigma_0$, the standard nesting assumption that goods within an industry are closer substitutes than goods across industries.

A shock confined to industry g is the worst case. An economy-wide regulation with regressive incidence and vanishing burden also raises other industries' price indices, directing expenditure back toward g and relaxing the condition.

Proof. Taking logs of profit gives

$$\ln \pi_j^g = \ln(Y/\sigma_g) + \ln S_g + (1 - \sigma_g)\big(\ln p_j - \ln P_g\big).$$

Differentiate at the top of the productivity distribution, where the burden vanishes, $d \ln \psi_j = 0$. With a constant markup, $d \ln p_j = d \ln \psi_j = 0$, so the within-industry term contributes $(\sigma_g - 1)\, d \ln P_g$.

The industry's log expenditure share is

$$\ln S_g = \ln \alpha_g + (1 - \sigma_0) \ln P_g - \ln \left(\sum_h \alpha_h P_h^{1-\sigma_0} \right).$$

For a shock confined to industry g, $d\ln P_h = 0$ for $h \neq g$. Differentiating therefore gives

$$\begin{aligned} d\ln S_g &= (1-\sigma_0)\, d\ln P_g - S_g(1-\sigma_0)\, d\ln P_g \\ &= (1-\sigma_0)(1-S_g)\, d\ln P_g. \end{aligned}$$

Combining the two contributions gives

$$d\ln \pi_j^g = \big[(\sigma_g - 1) + (1-\sigma_0)(1-S_g)\big]\, d\ln P_g.$$

The bracketed term is positive if and only if $\sigma_g - 1 > (\sigma_0 - 1)(1 - S_g)$. The condition $\sigma_g > \sigma_0$ is sufficient because

$$\sigma_g - 1 > \sigma_0 - 1 \geq (\sigma_0 - 1)(1 - S_g).$$

Under an economy-wide regulation, $d\ln P_h > 0$ for $h \neq g$. These terms enter $d\ln S_g$ with a positive sign, relaxing the condition. □

M Welfare with Homogeneous Goods: Derivations

This appendix derives the conditions under which total surplus increases in Cournot competition and in the dominant-firm model.

M.1 Cournot: proof of Proposition 7

Proposition 7 (When the homogeneous model reports a gain). *A marginal regulation that raises rivals' costs ($dc_f > 0$) and spares the leader changes total surplus by*

$$\frac{dTS}{dc_f} = \frac{n}{n+2}\big[q_L - (n+4)q_f\big].$$

Total surplus increases if and only if

$$\frac{c_f - c_L}{a - c_L} > \frac{n+3}{3n+8},$$

or, equivalently, if

$$s_L > s_L^*(n) \equiv \frac{n+4}{2(n+2)} \in \left(\tfrac{1}{2}, \tfrac{5}{6}\right].$$

Proof. Demand is $p = a - bQ$. The leader has marginal cost c_L, and each of the n rivals has marginal cost $c_f > c_L$. All $n+1$ firms choose quantities simultaneously. Each firm's

first-order condition is $p - c_i = b\,q_i$, giving the equilibrium

$$q_L = \frac{(a - c_L) + n(c_f - c_L)}{(n+2)b},$$
$$q_f = \frac{(a - c_L) - 2(c_f - c_L)}{(n+2)b},$$
$$p = \frac{a + c_L + n c_f}{n+2}.$$

Total surplus is $TS = \int_0^Q P(x)\,dx - c_L q_L - n c_f q_f$. Differentiating with respect to c_f and using $P(Q) = p$ gives the decomposition in Section 5.2:

$$\frac{dTS}{dc_f} = \sum_i (p - c_i)\frac{dq_i}{dc_f} - n q_f.$$

Equilibrium quantities respond according to

$$\frac{dq_L}{dc_f} = \frac{n}{(n+2)b}, \qquad \frac{dq_f}{dc_f} = -\frac{2}{(n+2)b}.$$

Substituting these responses and the first-order conditions $p - c_L = b q_L$ and $p - c_f = b q_f$ yields

$$\begin{aligned}\frac{dTS}{dc_f} &= b q_L \frac{n}{(n+2)b} - n b q_f \frac{2}{(n+2)b} - n q_f \\ &= \frac{n}{n+2}\big[q_L - (n+4) q_f\big].\end{aligned}$$

Substituting the equilibrium quantities into the bracket gives

$$q_L - (n+4) q_f = \frac{-(n+3)(a - c_L) + (3n+8)(c_f - c_L)}{(n+2)b}.$$

This expression is positive if and only if $(c_f - c_L)/(a - c_L) > (n+3)/(3n+8)$, establishing the first condition.

To express the condition in market shares, write total output as

$$Q = q_L + n q_f = \frac{(n+1)(a - c_L) - n(c_f - c_L)}{(n+2)b}.$$

The leader's share is therefore

$$s_L = \frac{(a - c_L) + n(c_f - c_L)}{(n+1)(a - c_L) - n(c_f - c_L)}.$$

This share is increasing in $(c_f - c_L)/(a - c_L)$ because the numerator increases and the denominator decreases. Writing $g \equiv (c_f - c_L)/(a - c_L)$ gives $s_L = (1 + ng)/\big((n+1) - ng\big)$. At the threshold $g^* = (n+3)/(3n+8)$, the numerator equals $(n+2)(n+4)/(3n+8)$ and the denominator equals $2(n+2)^2/(3n+8)$. Hence

$$s_L^*(n) = \frac{(n+2)(n+4)}{2(n+2)^2} = \frac{n+4}{2(n+2)}.$$

The threshold declines from $5/6$ at $n = 1$ toward $1/2$ as $n \to \infty$. Since the share is monotone in the cost gap, the two conditions are equivalent. □

M.2 The dominant firm

In the dominant-firm structure of Salop and Scheffman (1987), the leader chooses its quantity first. The n rivals then play Cournot among themselves, taking the leader's quantity as given. Write $A = a - c_L$ and $\Delta = c_f - c_L$. Equilibrium quantities are

$$q_L = \frac{A + n\Delta}{2b}, \qquad q_f = \frac{A - (n+2)\Delta}{2b(n+1)}.$$

The rivals' first-order condition is $p - c_f = bq_f$. For the leader, it is $p - c_L = bq_L/(n+1)$ because the leader internalizes the rivals' response.

Quantities respond according to

$$\frac{dq_L}{dc_f} = \frac{n}{2b}, \qquad \frac{dq_f}{dc_f} = -\frac{n+2}{2b(n+1)}.$$

Substituting into the same total-surplus decomposition gives

$$\begin{aligned}\frac{dTS}{dc_f} &= \frac{bq_L}{n+1} \cdot \frac{n}{2b} - nbq_f \cdot \frac{n+2}{2b(n+1)} - nq_f \\ &= \frac{n}{2(n+1)}\big[q_L - (3n+4)q_f\big].\end{aligned}$$

Total surplus therefore increases if and only if

$$\frac{c_f - c_L}{a - c_L} > \frac{2n+3}{4n^2 + 11n + 8},$$

or, equivalently, if

$$s_L > \frac{3n+4}{4(n+1)} \in \left(\tfrac{3}{4}, \tfrac{7}{8}\right].$$

The factor $n+1$ in the leader's margin means that a given share corresponds to a smaller

cost gap than under Cournot. The share threshold is therefore higher.

N Matching the Homogeneous Model

Section 5 fits the homogeneous model to a category's leader share and firm count. This appendix shows that a differentiated model can match the same moments. A nested demand system reproduces the homogeneous equilibrium's price, shares, and markups exactly, with every parameter determined by observables.

The construction applies to any equilibrium in which the industry sells at a common price p^*, a leader with marginal cost c_L holds share s_L, and n followers with marginal cost $c_f \in (c_L, p^*)$ hold the remaining share. The Cournot equilibrium in Section 5.1 is one example. The Stackelberg equilibrium below provides a second.

N.1 The homogeneous target: Stackelberg

Let the leader commit to its quantity first. The n followers then play Cournot among themselves, taking that quantity as given and facing the linear demand of Section 5. Write $A \equiv a - c_L$ and $\Delta \equiv c_f - c_L > 0$. Each follower chooses q_i to maximize $(p - c_f)q_i$, taking the quantities of the leader and the other followers as given. At the symmetric follower equilibrium,

$$q_f(q_L) = \frac{A - \Delta - bq_L}{b(n+1)}.$$

Substituting into demand gives the price faced by the leader:

$$p(q_L) = \frac{a + nc_f - bq_L}{n+1}.$$

Maximizing $(p(q_L) - c_L)q_L$ then yields

$$\begin{aligned} q_L &= \frac{A + n\Delta}{2b}, \\ p^* &= \frac{a + (n+1)c_L + nc_f}{2(n+1)}, \\ s_L &= \frac{(n+1)(A + n\Delta)}{(2n+1)A - n\Delta}. \end{aligned}$$

At $n = 0$, these expressions give the monopoly solution. At $\Delta = 0$, they give the symmetric Stackelberg outcome, with the leader holding $(n+1)/(2n+1) > 1/2$ of the market. The Lerner indices are $(p^* - c_L)/p^*$ for the leader and $(p^* - c_f)/p^*$ for each follower,

with a difference of Δ/p^*.

N.2 The differentiated model

The differentiated model nests the leader's variety and the followers' varieties in a two-tier CES system. The upper tier has elasticity σ_U and weights α_L, α_F over the leader's good and the follower composite. The follower composite is a CES aggregator with elasticity σ_F over the n follower varieties. Firms are atomistic in the sense of Section 2.1: each takes the relevant aggregator as given.

At symmetric follower prices p_f, the follower composite's price index is

$$P_F = n^{1/(1-\sigma_F)} p_f.$$

The CES pricing conditions are

$$p_L = \frac{\sigma_U}{\sigma_U - 1} c_L, \qquad p_f = \frac{\sigma_F}{\sigma_F - 1} c_f.$$

The leader's Lerner index is $1/\sigma_U$, and each follower's is $1/\sigma_F$.

N.3 An exactly determined calibration

Matching these Lerner indices to those of the homogeneous equilibrium determines both elasticities:

$$\sigma_U = \frac{p^*}{p^* - c_L}, \qquad \sigma_F = \frac{p^*}{p^* - c_f}.$$

At these elasticities, both prices equal p^*. Since $c_L < c_f < p^*$, both elasticities exceed one and $\sigma_U < \sigma_F$. The leader occupies the less elastic tier, where its higher markup offsets its lower cost. This allows firms with different marginal costs to charge the same price.

At a common price, expenditure shares equal quantity shares. Let e_L denote the leader's expenditure share. The upper-tier share equation is

$$\frac{e_L}{1 - e_L} = \left(\frac{\alpha_L}{\alpha_F}\right)^{\sigma_U} n^{-\frac{1-\sigma_U}{1-\sigma_F}} = \frac{s_L}{1 - s_L}.$$

Solving for the weight ratio gives

$$\frac{\alpha_L}{\alpha_F} = \left[\frac{s_L}{1 - s_L} n^{\frac{1-\sigma_U}{1-\sigma_F}}\right]^{1/\sigma_U}.$$

With equal costs, each follower receives $(1-s_L)/n$. Setting industry expenditure to $Y = p^*Q^*$

matches the scale of the homogeneous equilibrium.

Every parameter is determined by the observables $(p^*, c_L, c_f, s_L, n, Q^*)$; none remains free. The calibrated system reproduces the homogeneous equilibrium's price, every firm's share, and every firm's markup.

N.4 Where the match breaks

Now multiply the followers' cost c_f by $1 + \varepsilon > 1$, holding the calibrated parameters fixed. In the homogeneous model, the leader's price increases in strategic response to its followers' higher cost. In the calibrated differentiated model, the leader's price remains at $\sigma_U c_L/(\sigma_U - 1)$. As an atomistic firm, it does not adjust its price to a cost shock in the other tier, because the followers affect it only through the aggregator.

The followers' price index P_F increases, raising the leader's expenditure share. Reallocation proceeds in the same direction as in the homogeneous model, but differs in magnitude. The two models agree on every fitted moment at the baseline and diverge when the regulation arrives.

References

Abel-Koch, J. (2013). Endogenous trade policy with heterogeneous firms. Working paper, University of Mainz.

Andersen, D. C. (2018). Accounting for loss of variety and factor reallocations in the welfare cost of regulations. *Journal of Environmental Economics and Management 88*, 69–94.

Andreyeva, T., M. W. Long, and K. D. Brownell (2010). The impact of food prices on consumption: A systematic review of research on the price elasticity of demand for food. *American Journal of Public Health 100*(2), 216–222.

Anouliès, L. (2017). Heterogeneous firms and the environment: A cap-and-trade program. *Journal of Environmental Economics and Management 84*, 84–101.

Arkolakis, C., A. Costinot, and A. Rodríguez-Clare (2012). New trade models, same old gains? *American Economic Review 102*(1), 94–130.

Atkeson, A. and A. Burstein (2008). Pricing-to-market, trade costs, and international relative prices. *American Economic Review 98*(5), 1998–2031.

Autor, D., D. Dorn, L. F. Katz, C. Patterson, and J. Van Reenen (2020). The fall of the labor share and the rise of superstar firms. *Quarterly Journal of Economics 135*(2), 645–709.

Baccianti, C. and O. Schenker (2022). Cournot, pigou, and ricardo walk into a bar: Unilateral environmental policy and leakage with market power and firm heterogeneity. *Journal of the Association of Environmental and Resource Economists 9*(6), 1161–1195.

Baqaee, D. R. and E. Farhi (2020). Productivity and misallocation in general equilibrium. *Quarterly Journal of Economics 135*(1), 105–163.

Bartel, A. P. and L. G. Thomas (1985). Direct and indirect effects of regulation: A new look at OSHA's impact. *Journal of Law and Economics 28*(1), 1–25.

Bartel, A. P. and L. G. Thomas (1987). Predation through regulation: The wage and profit effects of the Occupational Safety and Health Administration and the Environmental Protection Agency. *Journal of Law and Economics 30*(2), 239–264.

Bartelsman, E., J. Haltiwanger, and S. Scarpetta (2013). Cross-country differences in productivity: The role of allocation and selection. *American Economic Review 103*(1), 305–334.

Becker, R. and V. Henderson (2000). Effects of air quality regulations on polluting industries. *Journal of Political Economy 108*(2), 379–421.

Berlingieri, G., S. Calligaris, and C. Criscuolo (2018). The productivity-wage premium: Does size still matter in a service economy? *AEA Papers and Proceedings 108*, 328–333.

Bloom, N., F. Guvenen, B. S. Smith, J. Song, and T. von Wachter (2018). The disappearing large-firm wage premium. *AEA Papers and Proceedings 108*, 317–322.

Brown, C. and J. Medoff (1989). The employer size-wage effect. *Journal of Political Economy 97*(5), 1027–1059.

Cai, D. and J. Li (2020). Pollution for sale: Firms' characteristics and lobbying outcome. *Environmental and Resource Economics 77*, 539–564.

Comin, D., D. Lashkari, and M. Mestieri (2021). Structural change with long-run income and price effects. *Econometrica 89*(1), 311–374.

Covarrubias, M., G. Gutiérrez, and T. Philippon (2020). From good to bad concentration? US industries over the past 30 years. *NBER Macroeconomics Annual 34*, 1–46.

Curtis, E. M. and J. M. Lee (2018). The reallocative and heterogeneous effects of cap-and-trade. *Economics Letters 172*, 93–96.

De Loecker, J., J. Eeckhout, and G. Unger (2020). The rise of market power and the macroeconomic implications. *Quarterly Journal of Economics 135*(2), 561–644.

Deaton, A. and J. Muellbauer (1980). An almost ideal demand system. *American Economic Review 70*(3), 312–326.

Demirer, M., D. J. Jiménez Hernández, D. Li, and S. Peng (2026). Data, privacy laws and firm production: Evidence from the GDPR. Working Paper 32146, National Bureau of Economic Research.

Dobkin, C., A. Finkelstein, R. Kluender, and M. J. Notowidigdo (2018). The economic consequences of hospital admissions. *American Economic Review 108*(2), 308–352.

Edmond, C., V. Midrigan, and D. Y. Xu (2023). How costly are markups? *Journal of Political Economy 131*(7), 1619–1675.

Fabra, N. and M. Reguant (2014). Pass-through of emissions costs in electricity markets. *American Economic Review 104*(9), 2872–2899.

Feenstra, R. C. (2003). A homothetic utility function for monopolistic competition models, without constant price elasticity. *Economics Letters 78*(1), 79–86.

Forslid, R., T. Okubo, and K. H. Ulltveit-Moe (2018). Why are firms that export cleaner? International trade, abatement and environmental emissions. *Journal of Environmental Economics and Management 91*, 166–183.

Fowlie, M., M. Reguant, and S. P. Ryan (2016). Market-based emissions regulation and industry dynamics. *Journal of Political Economy 124*(1), 249–302.

Fowlie, M. L. (2009). Incomplete environmental regulation, imperfect competition, and emissions leakage. *American Economic Journal: Economic Policy 1*(2), 72–112.

Freyaldenhoven, S., C. Hansen, J. Pérez Pérez, and J. M. Shapiro (2026). Visualization, identification, and estimation in the linear panel event-study design. In V. Chernozhukov, J. Hörner, E. La Ferrara, and I. Werning (Eds.), *Advances in Economics and Econometrics: Twelfth World Congress*, Volume 2, pp. 225–268. Cambridge University Press.

Gabaix, X. (2011). The granular origins of aggregate fluctuations. *Econometrica 79*(3), 733–772.

Grey, F. (2018). Corporate lobbying for environmental protection. *Journal of Environmental Economics and Management 90*, 23–40.

Harberger, A. C. (1971). Three basic postulates for applied welfare economics: An interpretive essay. *Journal of Economic Literature 9*(3), 785–797.

Helpman, E., O. Itskhoki, and S. Redding (2010). Inequality and unemployment in a global economy. *Econometrica 78*(4), 1239–1283.

Herrendorf, B., R. Rogerson, and Á. Valentinyi (2013). Two perspectives on preferences and structural transformation. *American Economic Review 103*(7), 2752–2789.

Hill, B. M. (1975). A simple general approach to inference about the tail of a distribution. *The Annals of Statistics 3*(5), 1163–1174.

Hopenhayn, H. A. (1992). Entry, exit, and firm dynamics in long run equilibrium. *Econometrica 60*(5), 1127–1150.

Hottman, C. J., S. J. Redding, and D. E. Weinstein (2016). Quantifying the sources of firm heterogeneity. *Quarterly Journal of Economics 131*(3), 1291–1364.

Hsieh, C.-T. and P. J. Klenow (2009). Misallocation and manufacturing TFP in China and India. *Quarterly Journal of Economics 124*(4), 1403–1448.

Huneeus, F. and I. S. Kim (2024). The effects of firms' lobbying on resource misallocation. Working Paper.

Hviid, M. and M. Olczak (2016). Raising rivals' fixed costs. *International Journal of the Economics of Business 23*(1), 1–26.

Kennard, A. (2020). The enemy of my enemy: When firms support climate change regulation. *International Organization 74*(2), 187–221.

Kimball, M. S. (1995). The quantitative analytics of the basic neomonetarist model. *Journal of Money, Credit and Banking 27*(4), 1241–1277.

Kwon, S. Y., Y. Ma, and K. Zimmermann (2024). 100 years of rising corporate concentration. *American Economic Review 114*(7), 2111–2140.

Landes, E. M. (1980). The effect of state maximum-hours laws on the employment of women in 1920. *Journal of Political Economy 88*(3), 476–494.

Macedoni, L. and A. Weinberger (2026). Lobbying for regulations: When big business says yes. Working Paper.

Maloney, M. T. and R. E. McCormick (1982). A positive theory of environmental quality regulation. *Journal of Law and Economics 25*(1), 99–123.

Marvel, H. P. (1977, October). Factory regulation: A reinterpretation of early english experience. *Journal of Law and Economics 20*(2), 379–402.

Matsuyama, K. and P. Ushchev (2017). Three alternative classes of flexible homothetic demand systems. SSRN Working Paper.

Matsuyama, K. and P. Ushchev (2020). Constant pass-through. CEPR Discussion Paper No. 15475 / SSRN Working Paper.

Melitz, M. J. (2003). The impact of trade on intra-industry reallocations and aggregate industry productivity. *Econometrica 71*(6), 1695–1725.

Melitz, M. J. and G. I. P. Ottaviano (2008). Market size, trade, and productivity. *The Review of Economic Studies 75*(1), 295–316.

Michaelis, P. (1994). Regulate us, please! On strategic lobbying in Cournot-Nash oligopoly. *Journal of Institutional and Theoretical Economics 150*(4), 693–709.

Muñoz-García, F. and S. B. Akhundjanov (2016). Can polluting firms favor regulation? *The B.E. Journal of Economic Analysis & Policy 16*(4).

Olley, G. S. and A. Pakes (1996). The dynamics of productivity in the telecommunications equipment industry. *Econometrica 64*(6), 1263–1297.

Oster, S. (1982). The strategic use of regulatory investment by industry sub-groups. *Economic Inquiry 20*(4), 604–618.

Pashigian, B. P. (1984). The effect of environmental regulation on optimal plant size and factor shares. *Journal of Law and Economics 27*(1), 1–28.

Peltzman, S. (1976). Toward a more general theory of regulation. *Journal of Law and Economics 19*(2), 211–240.

Rebeyrol, V. (2023). Protection without discrimination. *Economic Journal 133*(655), 2807–2827.

Salop, S. C. and D. T. Scheffman (1983). Raising rivals' costs. *American Economic Review 73*(2), 267–271.

Salop, S. C. and D. T. Scheffman (1987). Cost-raising strategies. *Journal of Industrial Economics 36*(1), 19–34.

Santos Silva, J. M. C. and S. Tenreyro (2006). The log of gravity. *Review of Economics and Statistics 88*(4), 641–658.

Stigler, G. J. (1971). The theory of economic regulation. *The Bell Journal of Economics and Management Science 2*(1), 3–21.

Syverson, C. (2011). What determines productivity? *Journal of Economic Literature 49*(2), 326–365.

Williamson, O. E. (1968). Wage rates as a barrier to entry: The Pennington case in perspective. *Quarterly Journal of Economics 82*(1), 85–116.

Zhelobodko, E., S. Kokovin, M. Parenti, and J.-F. Thisse (2012). Monopolistic competition: Beyond the constant elasticity of substitution. *Econometrica 80*(6), 2765–2784.